Inequality, Informality, and Optimal Progressivity*

Oscar Becerra
Universidad de los Andes – CEDE

Luigi-Maria Briglia
University of Tübingen

John León-Díaz
Inter-American Development Bank

Ralph Luetticke
University of Tübingen and the CEPR

Oscar Valencia
Inter-American Development Bank

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How should governments tax labor income when workers can shift into untaxed informal work? Combining Latin American household surveys with a heterogeneous-agent model of search, savings, and sectoral labor supply, we quantify the effects of progressivity in Mexico. Welfare-maximizing progressivity is over six times the current level, as gains from formalization among low earners initially outweigh the costs of evasion among high earners. The reform raises steady-state welfare by 4.3% of consumption and reduces informality by 5.4 percentage points. At equal revenue, progressivity delivers three to four times the welfare gain of higher labor- or capital-tax levels. Greater earnings inequality increases these gains while leaving optimal progressivity nearly unchanged.

JEL Codes: E26, H24, H26, O17, D31.

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1 Introduction

Should developing economies subsidize formal work through negative income taxes? Many OECD tax systems subsidize work at low earnings. Such policies are rare in economies with widespread informality, where much income falls outside the tax net. Subsidies tied to formal earnings do not directly reach workers employed only informally. Yet tax policy also changes where workers supply their labor. We ask how far progressive taxation can improve welfare and reduce informality, and whether changes in labor- or capital-tax levels can achieve similar gains.

We study this question with household survey evidence, a simple model of sectoral labor choice, and a quantitative heterogeneous-agent model calibrated to Mexico. We find substantial scope for redistribution through more progressive labor taxation. Subsidies to low formal earnings encourage low earners to work more hours formally. Higher marginal rates at the top finance these subsidies but also encourage high earners to shift hours into informality. Up to a point, the response at the bottom expands the tax base, allowing redistribution and formalization to reinforce one another.

The data show how access to formal jobs and returns to informal work vary across the income distribution. In nationally representative surveys for Brazil, Colombia, Mexico, and Peru, workers in lower-income households enter formal employment less often and leave it more often. Informality is highest at the bottom but remains common even in the top decile. Following the same workers across sectors in the Mexican panel, we estimate an elasticity of informal earnings with respect to formal earnings of approximately 0.8. Informal earnings thus rise less than one-for-one with formal earning power, so the earnings cost of moving into informality increases with ability. Informal earnings are also more volatile: the transitory variance of log earnings is about 2.4 times as large as in formal jobs. These findings discipline job access and sectoral labor choices and inform alternative calibrations of earnings risk.

A simple model isolates how job access and informal returns shape optimal progressivity. Households allocate labor between formal work, which is subject to unemployment risk, and informal work, which is always available but offers lower returns to ability. Informal work therefore provides income when formal earnings are lost. Subsidies to low formal earnings draw labor into the tax base—an *inclusion margin*. Higher marginal rates encourage high earners to shift labor into the untaxed sector—an *evasion margin*. Welfare initially rises with progressivity as inclusion gains dominate, then falls as evasion erodes the tax base. Optimal progressivity is 0.24, compared with about 0.60 without unemployment risk or informal work, holding the ability distribution and fiscal rule fixed.

The quantitative model adds productivity risk, savings, search frictions, and equilibrium changes in wages, vacancies, and asset prices. Job finding and separation determine who has a formal job. Households with a formal job choose how many hours to work in each sector, while savings provide insurance against earnings risk. We calibrate the model to Mexican earnings, wealth, consumption inequality, labor-market outcomes, and public finances. We choose benchmark progressivity to match consumption inequality. The resulting effective tax rates across income deciles are close to Mexican fiscal estimates that are not targeted. The quantitative analysis yields three main results.

Our first result is that Mexico can raise progressivity substantially while improving welfare and reducing informality. The welfare-maximizing progressivity parameter is 0.10, over six times the benchmark level of 0.015. The reform makes the marginal tax rate negative in the bottom decile of formal earnings and raises it from about 15% to 29% in the top decile. It generates a steady-state welfare gain of 4.3% in consumption equivalents, accounting for both consumption and leisure. The informal share of hours worked falls by 5.4 percentage points, consumption inequality falls by 3.2%, and tax revenue rises by about half a percentage point of GDP. Government spending remains fixed as a share of output, and changes in revenue net of spending are returned to households through the lump-sum fiscal instrument.

Up to the optimum, most formalization comes from households who already have a formal job. Those with lower productivity increase their formal labor supply, while those with higher productivity remain almost fully formal. The welfare-maximizing degree of progressivity also maximizes revenue as a share of GDP. Beyond this point, high-productivity workers shift into informal work, and welfare and revenue fall. Aggregate informality nevertheless continues to decline because vacancy creation expands access to formal jobs. Lower aggregate informality can therefore coexist with a shrinking tax base when high earners move income into informal work.

Our second result is that, compared with other tax instruments, progressivity delivers larger welfare gains at the same revenue increase while substantially reducing informality. At the revenue increase generated by the optimal progressivity reform, progressivity yields a welfare gain of 4.3% of consumption, compared with 1.4% from higher labor-tax levels and 1.2% from higher capital taxes. The welfare-maximizing labor-tax level yields a larger gain, 6.3%, but raises much more revenue and increases informality by almost 20 percentage points. These differences reflect how the instruments redistribute. Higher labor-tax levels finance transfers while reducing the return to formal work throughout the earnings distribution. Capital taxation redistributes from households with high asset income, with little effect on sectoral labor allocation. Progressivity instead raises

the return to formal work at the bottom and finances the subsidy through higher taxation at the top. Its welfare gains remain positive when transfers are fixed and government consumption adjusts. Under that budget rule, increases in labor- or capital-tax levels no longer improve welfare.

Our third result is that greater earnings inequality increases the welfare gains from progressivity while leaving its optimal level nearly unchanged. Higher productivity risk strengthens both margins. Low earners have more to gain from subsidies to formal work, while taxable income becomes more concentrated among high earners who can shift labor into informality. Between the low- and high-inequality economies, the welfare gain rises from 2.8% to 6.6% of consumption, while optimal progressivity moves only from 0.10 to 0.105. This differs from the usual prescription that greater uninsurable earnings risk calls for more progressivity (Heathcote et al., 2017, 2020). Here the stronger insurance motive is offset by a stronger evasion response.

Optimal progressivity remains well above its benchmark under alternative spending rules and calibrations. Holding government spending fixed in levels leaves optimal progressivity unchanged and yields a welfare gain of 3.0% of consumption. Recalibrating the model with transitory earnings risk, or with panel estimates of informal returns and earnings persistence, leaves optimal progressivity several times above the corresponding benchmark. These exercises show how earnings dynamics and the use of tax revenue shape the size of the welfare gains, while substantially higher progressivity remains welfare-improving.

Related Literature. Our paper connects optimal redistribution with incomplete markets to the macro-development literature on informality. Quantitative heterogeneous-agent models show how the tax code provides insurance and redistributes across households (Conesa and Krueger, 2006; Heathcote et al., 2017; Kindermann and Krueger, 2022), and how progressivity affects tax revenues (Holter et al., 2019). We adopt the parametric tax framework of Heathcote et al. (2014, 2017). Closest in this literature, Ferriere et al. (2023) show that implementable progressive tax systems can approach second-best allocations in fully formal economies. We make the tax base endogenous through formal–informal labor reallocation and compare progressivity with changes in labor- and capital-tax levels.

Doligalski and Rojas (2023) study optimal redistribution when workers can simultaneously supply formal and informal labor. Their Mirrleesian analysis shows how informal responses constrain redistribution. We study an implementable progressive schedule in a dynamic general-equilibrium environment with search frictions and precautionary saving. This framework allows us to quantify

how labor reallocation, self-insurance, and job availability jointly shape the gains from redistribution. On the consumption side, Bachas et al. (2024) show how informal purchases shape tax incidence and redistribution. We study labor informality as an endogenous response to the personal income tax schedule.

Our labor-market framework builds on dual-sector search models studying minimum wages and enforcement (Albrecht et al., 2009; Bosch and Esteban-Pretel, 2012), on-the-job search (Bobba et al., 2021), and wage sorting (Meghir et al., 2015). A complementary literature emphasizes firms' decisions to register and hire off the books (Ulyssea, 2018), compliance costs (Kleven et al., 2016), and firm organization over development (Engbom et al., 2025). Firm decisions and household labor choices jointly shape informality. We study the household margin directly affected by personal income taxation. Workers move between sectors throughout the income distribution, and households often combine both. We allow wages and formal job creation to respond in equilibrium, while taking firm registration decisions, corporate taxation, and enforcement as given. This framework quantifies how household labor choices shape redistribution, formalization, and the relative performance of tax instruments.

2 Empirical Evidence: Informality Across Incomes

We begin by documenting two features that shape the response to progressive taxation. The first is access to formal employment and the second is the payoff to informal work. Cross-country evidence shows that informality is highest among lower-income households, but remains sizable even at the top of the income distribution. Formal jobs are also harder to enter and less stable at the bottom. We then use the Mexican panel to measure how informal earnings vary with formal earning potential, separate persistent from transitory earnings risk, and document how workers combine formal and informal work. These moments discipline the structure and calibration of the quantitative model in Section 4.

2.1 The Data

We use nationally representative household surveys from Brazil (PNADC, 2024), Colombia (GEIH, 2023), Mexico (ENOE, 2024), and Peru (ENAHO, 2023). All four record labor earnings, labor-force status, informality identifiers, and demographics. Employment and unemployment follow each survey's standard definitions; informality follows country-specific criteria based on registration,

social-security coverage, and establishment characteristics. Appendix A describes in more detail these definitions, and Appendix F reports additional survey details and earnings and education patterns.

Our cross-country analysis covers individuals aged 15 and older in households with positive labor earnings. We construct country-specific deciles of household per-capita labor earnings, defined as the sum of household members' reported monthly labor earnings divided by household size. This measure excludes capital income and transfers. All estimates use survey weights. Labor earnings are highly unequal: the Gini ranges from 0.42 in Mexico to 0.58 in Colombia (Table F.1).

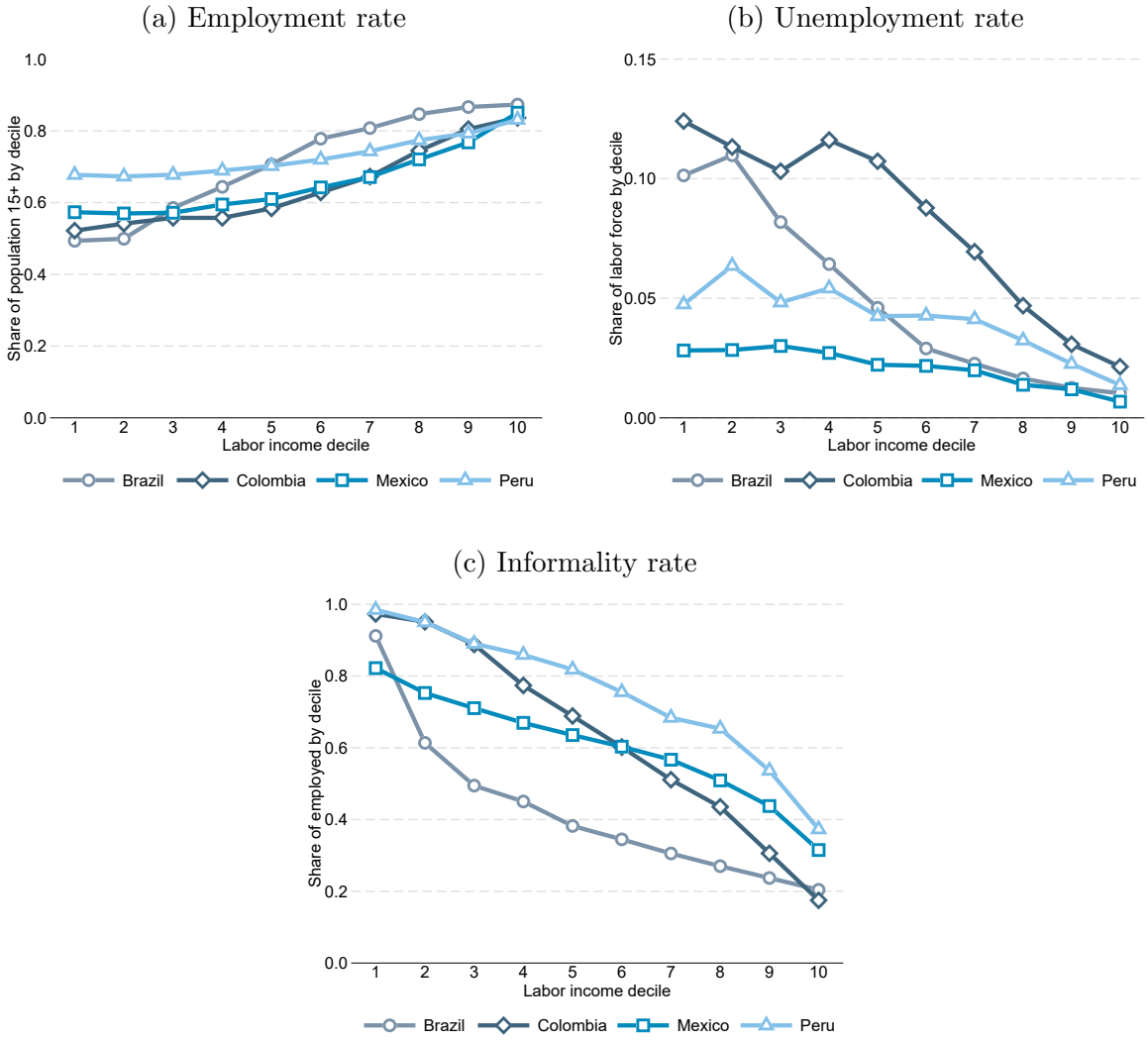
2.2 Labor Market Outcomes Across the Income Distribution

Figure 1 shows a clear income gradient in labor market outcomes. Employment rates rise with household income, while unemployment and informality generally decline. In Brazil, Colombia, and Mexico, employment is around 50% in the bottom decile and increases substantially toward the top. Differences in unemployment are particularly large between the bottom and top of the distribution in Brazil and Colombia. Informality follows an even clearer pattern. It is highest among lower income households, but remains sizable at the top of the distribution. Informality is therefore not confined to workers with the weakest labor market outcomes.

We use the panel structure of the Brazilian, Mexican, and Peruvian surveys, together with retrospective information from Colombia, to follow movements into and out of formal employment. Figure 2 shows that workers in lower income households are less likely to enter formal employment and more likely to leave it. Formal jobs are therefore both harder to obtain and less stable at the bottom of the distribution. At the same time, informality remains common among higher income workers, even though they have stronger attachment to formal employment. These patterns motivate a model in which search frictions determine access to formal jobs, while workers with formal access choose how to divide their labor between formal and informal activities. The transition profiles discipline the rates at which workers of different productivity levels find and lose formal jobs.

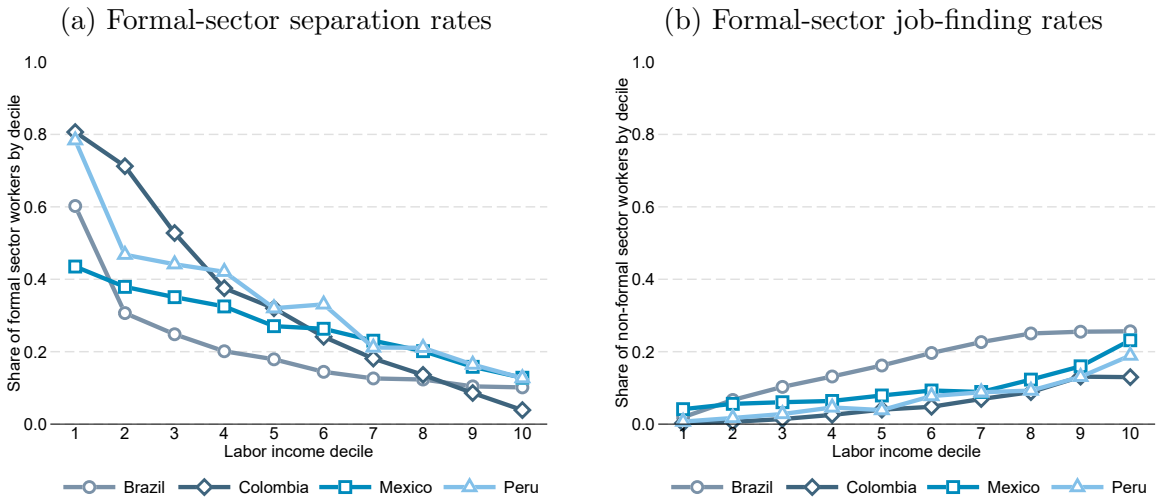
2.3 A Closer Look at Mexico: the ENOE Rotating Panel

The cross-country evidence shows that access to formal jobs varies substantially across the income distribution. We use the Mexican panel to study a second dimension, namely how the earnings cost of informal work varies with workers' earning power. The panel also allows us to distinguish



Notes: The sample covers individuals aged 15 and older in households with positive labor earnings. Deciles are constructed separately by country using household per-capita labor earnings. Survey years are 2024 for Brazil and Mexico and 2023 for Colombia and Peru. All estimates use survey weights. Informality follows country-specific definitions based on registration, social-security coverage, and establishment characteristics; see Appendix A.

Figure 1: Labor market indicators by labor income decile, 2023/24



Notes: Annual transition probabilities for individuals aged 15 and older in households with positive labor earnings. Separation means leaving formal employment for informal employment, unemployment, or nonparticipation; job finding means entering formal employment from these states. Deciles are constructed separately by country using household per-capita labor earnings. Survey years are 2024 for Brazil and Mexico and 2023 for Colombia and Peru. All estimates use survey weights. Country-specific formality definitions and survey details are given in Appendix A.

Figure 2: Annual formal-sector separation and job-finding rates, 2023/24

persistent from temporary earnings differences across sectors and to document the extent to which workers and households combine formal and informal employment. The ENOE interviews each household for five consecutive quarters. We link observations from 2021 to 2024, after the survey redesign, and obtain about 5.1 million person-quarter observations for 1.6 million individuals. The cross-sectional results above use the 2024 survey, while the analysis below uses the linked panel. Wage estimates are based on hourly earnings from the main job.

Informal returns to earning power. How much a worker gives up by moving work from a formal to an informal job, and how this loss varies with the worker's earning power, determines how easily earnings can leave the formal tax base. The empirical question is whether the formal-to-informal earnings penalty scales proportionally with earning power or rises with it.

We answer this with a simple measurement framework. Let h_{it} denote worker i's earning power at date t, measured in units of formal earnings, and let measured log hourly wages in the two sectors be

$$\ln W_{it}^{f} \;=\; \ln h_{it} + \nu_{it}^{f}, \qquad \ln W_{it}^{s} \;=\; a + \xi \ln h_{it} + \nu_{it}^{s},$$

where ν_{it}^{f} and ν_{it}^{s} are serially uncorrelated measurement errors. With $\xi = 1$ the formal-to-informal penalty is a constant percentage of earnings; with $\xi < 1$ it rises with earning power. For a worker

Table 1: Informal Returns to Formal Earning Power

Specification	Elasticity	Standard error
Salaried workers: baseline IV	0.840	0.027
Salaried workers: lag-matched IV	0.827	0.036
Including informal self-employment: joint IV	0.784	0.022

Notes: ENOE linked panel, 2021–2024, workers aged 25–60. Informal log hourly earnings are regressed on formal log hourly earnings, instrumented by a second formal observation of the same worker. The salaried sample contains 18,990 switchers and 102,102 wage triples, including 14,960 lag-matched triples. Lag matching places both wage observations equally far from the instrument. Earnings are residualized on observables. The last row adds informal self-employment (31,048 switchers) and jointly estimates the slope and intercept. Only baseline IV adjusts for mean reversion. Estimates use survey weights and worker-clustered bootstrap standard errors. Appendix A.3 gives the construction and identifying assumptions.

observed in informal work at date t and in formal work at date t', this implies

$$\ln W_{it}^{s} \;=\; a \;+\; \xi \,\ln W_{it'}^{f} \;+\; \nu_{it}^{s} - \xi\,\nu_{it'}^{f} + \xi\,(\ln h_{it} - \ln h_{it'})\,. \tag{1}$$

The slope ξ is the elasticity of informal to formal earnings for a given worker. The intercept a measures the level of informal relative to formal earnings, which reflects both sector-wide pay differences and the productivity of informal work; since it depends on the units of measurement, we report it below as a wage ratio.

We estimate equation (1) on workers observed in both sectors, which separates differences in returns from differences in worker composition (Gibbons et al., 2005; Lemieux, 1998). Measurement error in the formal wage biases ordinary least squares toward zero (Bound et al., 2001): OLS yields an elasticity of 0.36. We therefore instrument the formal log wage with a second formal observation of the same worker, which removes attenuation from serially uncorrelated measurement error (Griliches and Hausman, 1986). The sample requires at least one informal and two formal observations within the five interviews. Earning power can also change between t and t': the baseline estimate adjusts for mean reversion using estimated quarterly persistence, and a lag-matched sample places both wage observations equally far from the instrument, so no adjustment is required. Identification requires that workers do not select into switching on innovations to their earning power around the switch. Appendix A.3 describes the details.

Table 1 reports an elasticity of approximately 0.8, precisely estimated below one. Among these switchers, informal earnings rise less than one-for-one with formal earning power, so the penalty for informal work is not proportional: it rises by about 0.16–0.22 log points for every log point of

formal earning power. The lag-matched estimate is close to the baseline estimate, while including informal self-employment gives a somewhat lower elasticity. Appendix Table A.1 reports alternative wage concepts and informality definitions; the estimates range from 0.75 to 0.87. The joint slope–intercept specification for salaried workers with lag-matched observations also pins down the level of the penalty: an informal-to-formal wage ratio of 0.92 at mean formal earnings.

The same relationship holds across the formal wage distribution, in a comparison that uses different variation and no instrument. Relative to workers who start from the same formal-wage decile and remain formal, workers moving into informality lose 0.07–0.10 log points of earnings one quarter later in the second through sixth deciles, rising to 0.35 log points in the top decile. The loss widens mostly in the upper half of the distribution, and its overall widening is of the order the estimated elasticity implies (Appendix A.3). Workers with greater formal earning power give up more by moving into informal work.[1]

Persistent and transitory earnings variance. Informal earnings are more dispersed than formal earnings, and roughly half of this difference is transitory. We split each sector's cross-sectional variance of log earnings into persistent and transitory components. Autocovariances at lags one and beyond identify the persistent component under serially uncorrelated measurement error. Log-earnings variance is 0.674 in informal work and 0.325 in formal work. The gap of 0.349 consists of 0.169 in persistent differences and 0.180 in transitory fluctuations. Transitory variance is 0.313 in informal work and 0.132 in formal work, about 2.4 times as large. The distinction matters because households can use savings to cushion temporary earnings losses more readily than persistent ones. Section 4 targets the persistent part of the gap, and Appendix C adds the sectoral transitory variances in a robustness calibration.

Combining formal and informal work. In the ENOE panel, 2.4% of employed workers aged 25–60 hold jobs in both sectors, almost always a formal main job and an informal second job. Across household members, sector mixing is more common: 22.8% of working households have members in both sectors, rising to 39.9% among households with at least two employed members.[2] These

[1] Earlier panel studies for Mexico find formal–informal wage gaps that rise with education (Gong and van Soest, 2002) and, net of a worker fixed effect common across sectors, salaried penalties that shrink in quarters when workers earn above their own average (Bargain and Kwenda, 2014).

[2] Individual mixing is measured by the sector of the main and second jobs; the second job's sector is observed only in first-quarter waves. Household mixing uses all employed members aged 15 and older, classified by their main job, and includes households with at least one employed member. Both measures use survey weights from the 2021–2024 ENOE panel.

patterns distinguish labor allocation within workers from specialization across household members. Our single-earner model studies the within-worker margin, allowing workers with access to formal jobs to allocate hours between sectors.

3 A Simple Model of Progressive Taxation with Informal Work

This section presents a static model to ilustrate how progressive taxation affects inequality and labor allocation in economies with large informal sectors. Heterogeneous households choose total labor supply and divide their labor between salaried formal work and informal self-employment. Formal jobs offer higher returns to ability but expose households to unemployment risk, while informal work provides an always available source of income. The model captures both subsistence entrepreneurship and tax avoidance (Schoar, 2009).

The government taxes only formal earnings. Progressivity changes the return to formal work differently across the income distribution. At the bottom, negative income taxes make formal work more attractive and encourage low earners to shift labor into the formal sector. At the top, higher marginal rates make untaxed work more attractive and encourage the opposite reallocation. We refer to these responses as the *inclusion* and *evasion* margins. Their balance determines how progressivity affects informality, redistribution, and welfare.

Households. The economy consists of a continuum of households endowed with exogenous ability h, drawn from a log-normal distribution, $\ln h \sim \mathcal{N}(-\sigma_h^2/2, \sigma_h^2)$. Average ability is normalized to one. At the beginning of the period, households choose total labor supply $n \in [0,1]$ and the fraction $\varpi \in [0,1]$ allocated to formal work. The remaining fraction is allocated to informal self-employment. Formal labor earns $w^f hn\varpi$, but with probability q the household does not find a job and receives no formal earnings. This risk is common across ability levels and does not depend on ϖ. Allocating more time to formality changes earnings if a job is found, but leaves the probability of finding one unchanged.[3]

Informal work faces no matching frictions and earns $w^s \Xi h^\xi n(1-\varpi)$, where $\Xi < 1$ and $\xi < 1$. Earnings increase with ability in both sectors, but more slowly in informal work. Low-ability households therefore have a smaller opportunity cost of forgoing formal work, while high-ability households have more to gain from formal employment. Informality in this model refers to self-

[3]This probability differs from the vacancy-filling rate $q^v(\theta)$ in the quantitative model of Section 4.

employment, although informal salaried jobs are also widespread in developing economies.[4]

The government taxes formal earnings y through the schedule (Benabou, 2002; Ferriere et al., 2023; Heathcote et al., 2017)

$$T(y) = y - \tau^L y^{1-\tau^P}. \tag{2}$$

Here τ^P controls progressivity and τ^L scales after-tax formal earnings. Without progressivity, the proportional tax rate is $1 - \tau^L$; lowering τ^L raises the tax burden. With positive progressivity, the schedule allows negative income taxes at low earnings and positive taxes at higher earnings. Informal earnings are untaxed, reflecting the difficulty of observing income from informal work (Banerjee et al., 2024). Households may also receive a lump-sum transfer τ^T regardless of employment status.

Negative income taxes are paid only when households earn formal income. Redistribution through the tax schedule therefore depends on formal work. By shifting labor toward the formal sector, a household can increase formal earnings and gain access to the subsidy. The lump-sum transfer works differently. It is received regardless of employment status and can therefore reach households that work entirely informally.

Given ability h, households choose consumption and labor supply to maximize expected utility

$$u(c^f, c^s) = (1-q)\ln c^f + q \ln c^s - \frac{n^{1+\varphi}}{1+\varphi}, \tag{3}$$

where φ governs the curvature of labor disutility. Consumption satisfies the budgets when households find and do not find a formal job,

$$c^f = w^s \Xi h^\xi n(1-\varpi) + \tau^L (w^f h n \varpi)^{1-\tau^P} + \tau^T, \tag{4}$$

$$c^s = w^s \Xi h^\xi n(1-\varpi) + \tau^T. \tag{5}$$

Labor is allocated before formal job uncertainty is resolved. Informal earnings are available in both states, so informal work insures households against the loss of formal earnings. A household that works only informally has the same consumption in both states, but forgoes any gain from access to formal employment.

Production. A competitive firm combines aggregate formal and informal labor through the CES technology

$$Y = \left[(L^f)^{\frac{\epsilon-1}{\epsilon}} + (L^s)^{\frac{\epsilon-1}{\epsilon}} \right]^{\frac{\epsilon}{\epsilon-1}}, \tag{6}$$

[4]Labor informality extends beyond self-employment (Ulyssea, 2018). Much salaried informal employment is concentrated in small unregistered firms operated by self-employed individuals (La Porta and Shleifer, 2014).

where ϵ is the elasticity of substitution. Wages equal marginal products. The static model isolates the role of household skills by giving the two sectors the same aggregate productivity. Reallocation changes the relative scarcity of formal and informal labor and therefore their wages. These wage changes feed back into households' allocation choices. The quantitative model also allows sector-specific productivity.

Government. The government balances its budget, using formal income taxes to finance spending G and transfers τ^T. In the experiments below, transfers are zero and τ^L adjusts so that spending remains 15% of output in every simulation. Changes in progressivity therefore alter the shape of the tax schedule while preserving the government's share of output. In the quantitative progressivity experiment, the spending share and τ^L are held fixed, while τ^T adjusts to return additional revenue or collect any shortfall. Appendix D.2 examines alternative budget closures. The government budget and equilibrium conditions for the static model appear in Appendix E.

3.1 Formal and Informal Labor Allocation

Households compare the return to formal work with its unemployment risk and the earnings forgone in informal work. At an interior allocation, their choice satisfies

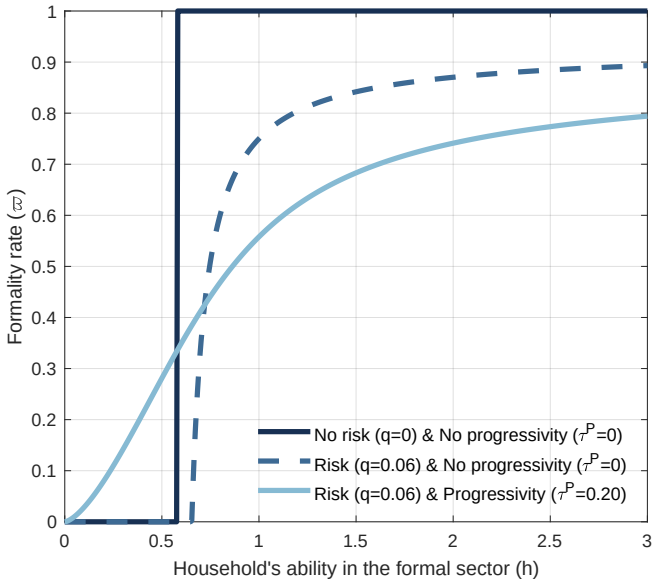
$$(1-q)\frac{(1-\tau^P)\tau^L(w^f hn)^{1-\tau^P}\varpi^{-\tau^P} - w^s \Xi h^\xi n}{c^f} = q\frac{w^s \Xi h^\xi n}{c^s}. \tag{7}$$

The left-hand side is the expected marginal utility gain from reallocating labor to formality when a job is found. The right-hand side is the expected marginal utility cost of the informal earnings forgone when no formal job is found. Because consumption is lower without a formal job, households value informal earnings particularly highly in that state. Their allocation therefore depends on both relative earnings and the insurance provided by informal work.

Without progressivity, low-productivity households choose only informal work when

$$h < \left(\frac{w^s \Xi}{(1-q)\tau^L w^f}\right)^{\frac{1}{1-\xi}}.$$

At given wages, higher unemployment risk or a higher proportional labor tax raises this cutoff. More households then specialize in informal work even when formal employment may be available. Households above the cutoff can still allocate some labor to informality as insurance against unemployment. The model thus captures both exclusion from formal work and voluntary informal



Notes: Households choose formal labor share ϖ before job availability is known; q is the probability of no formal job and τ^P controls tax progressivity. Wages and the tax scale are held fixed, and lump-sum transfers are zero.

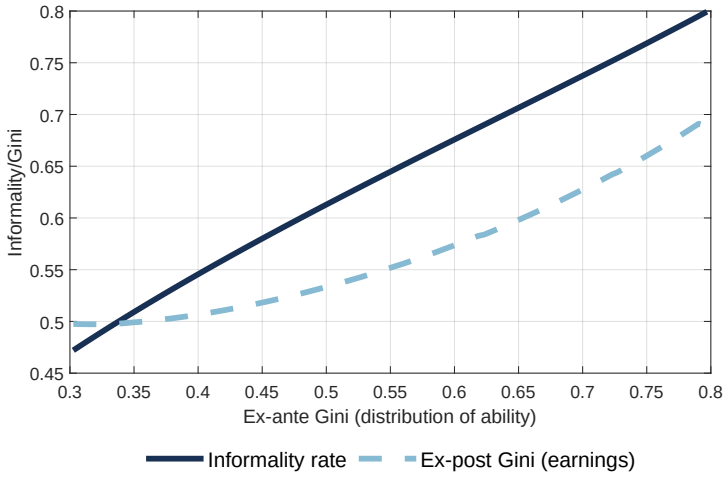
Figure 3: Formal-sector labor supply (ϖ)

employment (La Porta and Shleifer, 2014; Perry et al., 2007; Ulyssea, 2020).

Informality can arise even without taxes or unemployment risk. Households whose formal earnings per hour are below informal earnings, $w^f h < w^s \Xi h^\xi$, choose informal work. Taxes and unemployment risk expand this group by reducing the expected gain from formality. A subsidy to low formal earnings works in the opposite direction.

Figure 3 illustrates these choices across the ability distribution. Without unemployment risk or progressivity, households specialize in the sector with the higher return: low-ability households work informally and high-ability households work formally. Introducing unemployment risk changes the high-ability allocation. These households retain some informal work to insure against not finding a formal job, while low-ability households remain informal. In the illustrated economy, the formal share of labor still rises with ability, reflecting the increasing earnings advantage of formal work.

Progressivity affects low- and high-ability households differently. Negative income taxes encourage low-ability households to allocate part of their labor to formal work, strengthening the inclusion margin. High-ability households face higher marginal rates and shift labor toward untaxed work, strengthening the evasion margin. Informality therefore need not rise uniformly when taxation becomes more progressive. Its aggregate response depends on whether the increase in formal work



Notes: Ability dispersion increases while mean ability remains one. Tax progressivity and lump-sum transfers are zero. The tax scale adjusts in each static-model equilibrium to finance government spending equal to 15% of output; other structural parameters are unchanged.

Figure 4: Inequality and informality

among low earners outweighs the reduction among high earners.[5]

The figure holds wages fixed to isolate households' allocation decisions.[6] In the policy experiments that follow, wages adjust with aggregate formal and informal labor. These changes in relative returns affect households' labor choices, the formal tax base, and the resources available for consumption.

3.2 Inequality and Informality

The distribution of ability affects both the demand for redistribution and the prevalence of informal work. Greater dispersion places more households at low levels of ability, where the earnings advantage of formal employment is smaller, while also increasing inequality across households. The same underlying inequality can therefore affect both sides of the policy problem.

Figure 4 increases the dispersion of ability, σ_h, while holding mean ability at one through $\ln h \sim \mathcal{N}(-\sigma_h^2/2, \sigma_h^2)$. We set $q = 0.06$, $\tau^P = \tau^T = 0$, $\Xi = \xi = 0.5$, and $\epsilon = 7.65$, and adjust τ^L to finance spending of 15% of output.

A more dispersed ability distribution increases informality by placing more households at low

[5] We abstract from non-monetary job attributes, including flexibility and independence in self-employment (Maloney, 2004) and mandated formal-sector benefits (Levy, 2008). These could enter the allocation condition through additional preference terms (Galiani and Weinschelbaum, 2012; Gerard and Gonzaga, 2021).

[6] For this illustration, $w^f = 1.04$, $w^s = 1.26$, $\tau^L = 0.8$, and $\Xi = \xi = 0.5$.

productivity, where informal work is relatively attractive. This effect dominates the adjustment in relative wages. Informal work also compresses observed earnings differences. As the ability Gini rises from 0.30 to 0.80, the earnings Gini rises from about 0.50 to 0.70, substantially less than the increase in underlying ability inequality. These results are consistent with the positive association between inequality and informality documented in the literature (Chong and Gradstein, 2007; Rosser et al., 2000); Appendix B reports the corresponding cross-country evidence.

3.3 Welfare Implications

Fiscal policy changes consumption inequality directly through taxes and indirectly through labor reallocation and equilibrium wages. To assess their effects jointly, we solve for equilibrium at each value of progressivity. We set unemployment risk to $q = 0.06$, the elasticity of substitution to $\epsilon = 7.65$ (Leyva and Urrutia, 2020), informal productivity parameters to $\Xi = \xi = 0.5$, and log-ability dispersion to $\sigma_h = 1$. As above, τ^L adjusts to finance spending equal to 15% of output without lump-sum transfers. Without progressivity, informality is 62.6% and the earnings Gini is 0.54.

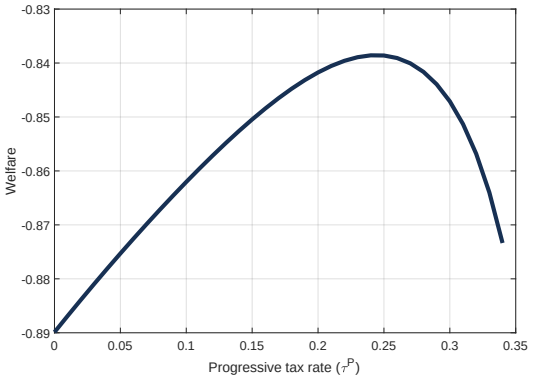
We evaluate welfare using households' expected utility, including the disutility of total labor. Formalization is not itself an objective of the welfare calculation. It matters only through its effects on earnings, consumption, labor effort, and the resources available across household employment states.

Figure 5 shows hump-shaped welfare and U-shaped informality. Starting from low progressivity, negative income taxes induce low-ability households to allocate more labor to formal work. This inclusion response initially outweighs the reduction in formal labor among high-ability households. The resulting gains in earnings and consumption at the bottom, together with greater compression of the earnings distribution, raise welfare. Redistribution and formalization therefore move together over this range of the tax schedule.

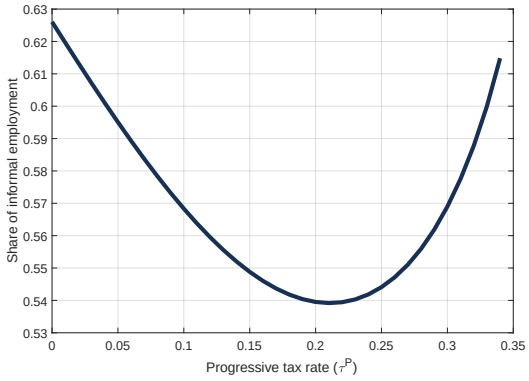
Beyond a threshold, the pattern reverses. Higher marginal rates induce high-ability households to shift more labor into informal work. The evasion margin eventually dominates, shrinking the formal tax base and aggregate output. Welfare peaks at moderate progressivity, where the gains from redistribution and inclusion remain large relative to these costs. In this simulation, the welfare-maximizing degree of progressivity is $\tau^P = 0.24$. Most labor adjustment occurs across sectors, while total labor supply changes relatively little, as shown in Appendix E.

Panel (c) compares welfare in our model with welfare in an economy without unemployment risk

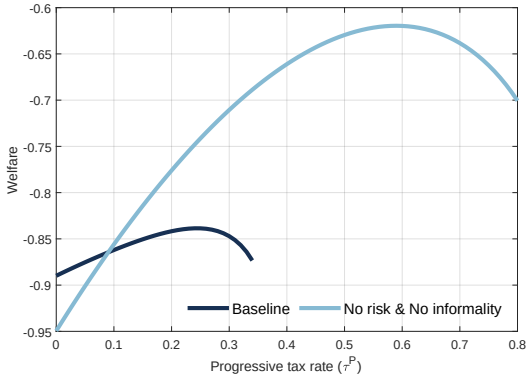
(a) Welfare (utilitarian)



(b) Informality rate



(c) Welfare comparison: no unemployment risk or informality



Notes: Welfare is population-average expected utility, including labor disutility, in levels. Transfers are zero, and the tax scale adjusts to finance government spending of 15% of output. The two economies in panel (c) share the same ability distribution and fiscal rule; the comparison removes both unemployment risk and informal work.

Figure 5: Progressivity, welfare, and informality

or informality, as in Ferriere et al. (2023). Under the same ability distribution, optimal progressivity in that benchmark is about 0.60, substantially above the value of 0.24 in the model with informality. The comparison removes both unemployment risk and the option to work informally, so it should not he interpreted as isolating either force separately. It does show that when workers can move labor outside the tax base, high marginal rates carry an additional cost that limits the scope for redistribution.

Appendix D.8 shows how the results vary with unemployment risk and public financing needs. Higher risk and greater financing needs each lower optimal progressivity. The scope for redistribution therefore depends both on the outside option available to households and on the amount of revenue the government must raise.

These results highlight why progressivity and the tax level have different effects on formal work. A higher proportional labor tax lowers the return to formal work across the distribution. Progressivity instead raises formal returns at the bottom while lowering them at the top, allowing redistribution to encourage formality over part of the policy range. The quantitative model asks how far this mechanism operates in Mexico and whether changes in labor or capital tax levels can achieve the same combination of redistribution and formalization.

4 A Quantitative Heterogeneous-Agent Model with Informality

The static model shows that progressivity can increase formal labor at the bottom of the distribution while stronger marginal incentives can encourage informality at the top. It also shows that informal work can provide insurance when formal employment is uncertain. The quantitative question is how important these forces remain once we account for the constraints households face and for the responses of firms, prices, and the government budget.

This section develops a dynamic heterogeneous agent model to address these questions. We first separate access to formal employment from the allocation of labor conditional on that access. Job finding and separation determine whether a household can work formally, while households with a formal job choose how much labor to supply in each sector. This distinction allows us to assess how much of the response to progressivity comes from workers reallocating hours and how much operates through changes in formal employment opportunities.

We then allow households to save against persistent earnings risk and firms to adjust vacancy creation in response to the value of formal labor. Sectoral reallocation affects wages, output, productivity, and tax revenue, so the welfare effects of progressivity depend not only on redistribution

but also on the resources available to finance it. The model is calibrated to Mexican evidence on earnings, employment transitions, inequality, and public finances, and is used to compare greater progressivity with alternative changes in labor and capital taxation.

4.1 Taxes and Transfers

The government levies a consumption tax τ^c, a proportional capital-income tax τ^k, and a flat tax τ^{profit} on the profit income of firm owners. Formal labor earnings face the progressive tax schedule introduced in Section 3. Let y denote gross formal labor earnings and $T(y)$ the associated tax liability. The parameter τ^L determines the level of taxation and τ^P its progressivity. We express the schedule relative to average gross formal earnings in the benchmark equilibrium, $\bar{y}$:

$$T(y) = y - \tau^L \left(y/\bar{y}\right)^{1-\tau^P} \bar{y}. \tag{8}$$

Expressing the schedule in relative earnings makes it independent of the units in which earnings are measured. Each reform uses the same anchor, given by average formal earnings in the benchmark.[7] Tax revenues finance government consumption G and a uniform lump-sum instrument τ^T. The government balances its budget every period.

4.2 Production

Following Leyva and Urrutia (2020), production combines three technologies. Final output uses capital and intermediate inputs. Limited managerial capacity, captured by a "span-of-control" constraint, creates decreasing returns at the firm level and generates positive profits. The intermediate input combines formal and informal production, which are imperfect substitutes. Within each sector, output is linear in labor and uses no other inputs. This structure allows sectoral labor reallocation to affect relative wages and measured TFP.

[7] In our calibration units are chosen so that average gross formal earnings equal one at the benchmark, so the schedule coincides with the familiar level form $T(y) = y - \tau^L y^{1-\tau^P}$ at the benchmark and, with the anchor held fixed, in the policy experiments. Writing the schedule in relative earnings is also what makes estimated (τ^L, τ^P) pairs comparable across countries and over time.

4.2.1 Final Good Production

A representative firm combines capital K and intermediate inputs H according to

$$Y = A\left(K^{\alpha}H^{1-\alpha}\right)^{\nu}, \quad (\alpha,\nu) \in (0,1)^2, \tag{9}$$

where Y is final output, A is aggregate productivity, α governs the capital intensity of production. The income shares of capital and the intermediate composite are $\alpha\nu$ and $(1-\alpha)\nu$, respectively. The parameter ν captures the span-of-control limitation. With $\nu < 1$, decreasing returns at the firm level generate positive profits, which are distributed to household owners.

Let p^f denote the price of formal intermediate output, w^s the return per efficiency unit of informal labor, and w^f the wage per efficiency unit of formal labor. Competitive factor prices equal marginal products. Appendix G.1 gives the factor-price equations and the aggregate production function.

4.2.2 Intermediate Goods Production

The intermediate input combines formal and informal output under imperfect substitutability. Let L^f and L^s denote formal and informal labor, respectively, measured in efficiency units. Each efficiency unit produces Ω units of output in the formal sector and $\varkappa$ units in the informal sector. These sectoral outputs enter a CES aggregator, whose elasticity of substitution ε determines how readily production can substitute one for the other:

$$H = \left[(\Omega L^f)^{\frac{\varepsilon-1}{\varepsilon}} + (\varkappa L^s)^{\frac{\varepsilon-1}{\varepsilon}}\right]^{\frac{\varepsilon}{\varepsilon-1}}. \tag{10}$$

Reallocating labor between the two technologies changes measured TFP through the composition of production. Let $\lambda^s = L^s/(L^f + L^s)$ denote the informal share of labor in efficiency units. Over the relevant range, a higher λ^s lowers measured TFP. The elasticity ε governs the size of this effect. Appendix G.1 gives the corresponding expression, and Section 5 reports its magnitude in the policy experiments.

4.3 Labor Market

Formal employment is subject to search and matching frictions, while the informal sector is perfectly competitive. Let $F \in \{0,1\}$ denote the formal employment status, with $F = 1$ indicating access to

a formal job. Employment status evolves through job finding and separation, so access to formal employment depends on market conditions and differs across workers. As in the static model, informal work is always available. It provides income when formal employment is unavailable and allows households with a formal job to shift hours outside the tax base. Earnings therefore depend on productivity, access to formal employment, the allocation of hours across sectors, and the nonlinear tax schedule.

Formal Labor Market. Workers without a formal job, $F = 0$, search for formal employment while working informally. Let I denote the mass of searching workers and V the number of vacancies posted by firms. A constant returns to scale matching function determines the number of matches $M(I, V)$:

$$M(I, V) = \chi \, I^{\psi} V^{1-\psi}, \tag{11}$$

$$f(\theta) = \frac{M}{I} = \chi \, \theta^{1-\psi}, \tag{12}$$

$$q^{v}(\theta) = \frac{M}{V} = \chi \, \theta^{-\psi}, \tag{13}$$

where χ is matching efficiency, ψ is the elasticity of matches with respect to searching workers, and $\theta \equiv V/I$ is labor market tightness. The functions $f(\theta)$ and $q^{v}(\theta)$ are the job finding rate for workers and the job filling rate for firms.

4.4 Households

A continuum of infinitely lived households derives utility from consumption and leisure. Let c denote consumption, a current asset holdings, a' next period's assets, and ℓ total hours worked. Households save in a risk-free asset that pays the equilibrium interest rate r. This rate is constant within each steady state, and interest income is subject to the capital income tax τ^{k}. Households also pay the consumption tax τ^{c} and the progressive tax on formal labor income in (8). They receive lump-sum payment τ^{T}, which can be positive or negative. The future is discounted at rate β.

Households with access to formal employment may work in both sectors. For these households, $\kappa \in [0, 1]$ denotes the share of total hours supplied formally, so formal and informal hours are $\kappa\ell$ and $(1 - \kappa)\ell$. Households choose consumption, savings, and hours in each sector to maximize expected lifetime utility.

Idiosyncratic Risk. Labor productivity follows a log-AR(1) process:

$$\ln h' = \rho \ln h + \sigma_h \epsilon_h, \qquad \epsilon_h \overset{\text{iid}}{\sim} \mathcal{N}(\mu_h, 1). \tag{14}$$

where ρ governs persistence, σ_h is the standard deviation of innovations, and ϵ_h is the innovation to productivity.[8] Households face uncertainty about labor productivity, access to formal employment, and firm ownership

Sector-Specific Worker Productivity. Productivity differs across sectors at both the technology and worker levels. The parameters Ω and $\varkappa$ describe sectoral technologies, as defined in Section 4.2.2. At the worker level, formal efficiency units equal potential productivity, $h^f = h$, while informal efficiency units are $h^s = \Xi h^\xi$. The parameters Ξ determines the level of informal worker productivity, while ξ determines how informal productivity varies with potential productivity. When $\xi < 1$, differences in worker productivity translate into smaller proportional differences in informal efficiency units.

A value below one can capture limits on expanding informal activity, including restricted access to capital, customers, and enforceable contracts. In the notation of equation (1), ξ is the elasticity of informal to formal earnings estimated in Section 2.3 among sector switchers, about 0.8. In the baseline, we calibrate ξ jointly with the other informal productivity parameters using sectoral earnings moments. Appendix C considers a calibration that fixes ξ at the panel estimate.

Risk across sectors. A common persistent productivity shock affects both sectors through $h^f = h$ and $h^s = \Xi h^\xi$. When $\xi < 1$, informal productivity responds less than proportionally to the common shock. With the calibrated value of ξ close to one, this difference is quantitatively small. Formal employment also faces productivity-specific job finding and separation rates, while informal work can be taken up immediately and at any intensity. Informal work therefore provides income after a formal-job loss while remaining exposed to productivity risk.[9] Lower informal returns to productivity also increase the earnings cost of moving into informal work for high-productivity

[8] As in the static model, the innovation mean parameter μ_h is set to satisfy the steady-state normalization condition $\mathbb{E}[h] = 1$.

[9] Gomes et al. (2020) document greater earnings volatility among informal workers and large earnings changes around formal and informal transitions in Brazil. Engbom et al. (2022) similarly find that earnings changes are substantially more dispersed among informal than formal workers, particularly among sector switchers. These findings are also consistent with our ENOE evidence in Section 2.3, where the estimated transitory component of earnings variance is 2.4 times larger in informal employment.

households. These returns and formal employment risk jointly shape labor allocation across sectors.[10]

Skill-Dependent Transition Rates. The average job-finding rate is $f(\theta)$, while the average separation rate $\bar{\iota}$ is exogenous. Both transition rates vary across productivity types in order to reproduce the systematic differences observed across the labor income distribution.

Each productivity type $h_i \in \{h_1, \ldots, h_{n_h}\}$ has a pair of transition parameters $\{\Delta f_{h_i}, \iota_{h_i}\}$. The term Δf_{h_i} is the type-specific deviation from the equilibrium job-finding rate, so the rate for type h_i becomes $f_{h_i} = f(\theta) + \Delta f_{h_i}$. The parameter ι_{h_i} is the corresponding separation rate. We calibrate the arrays $\{\Delta f_{h_i}\}$ and $\{\iota_{h_i}\}$ to the observed job finding and separation profiles across labor income deciles. Access to formal employment vary with productivity, while the average job-finding rate continues to responds to equilibrium vacancy posting.[11]

Employment status is known when households choose hours and savings. Thus, F determines access to a formal job, while κ determines the formal share of hours conditional on that access. Households with $F = 0$ can work only informally.

Risky Firm-Ownership Status. Only workers at the highest point on the productivity grid who have a formal job can enter the firm ownership state, with probability π^{up}. Following Castaneda et al. (2003), this state has a shorter expected duration than the other idiosyncratic states. Its duration is governed by the exit probability π^{fall}. On exit, the household moves to a productivity level drawn from an exponential distribution over the grid, with rate π^{unlucky}.[12]

4.4.1 Informally Employed Households' Problem

Households without a formal job can work as informal self-employed workers. They choose hours ℓ in a perfectly competitive market and receive w^s per efficiency unit of informal labor. They may enter formal employment next period. Let $V^s(a, h)$ and $V^f(a, h)$ denote the value functions of households without and with formal employment, respectively. The problem of a household without

[10] Informal jobs are not subject to destruction shocks in the model. Informal workers face risk through stochastic efficiency units. Empirically, informal labor markets display larger worker flows than formal ones (Donovan et al., 2023; Meghir et al., 2015). The baseline abstracts from separate transitory earnings risk. Appendix C reports a calibration with sector-specific transitory risk estimated from the ENOE panel.

[11] Worker productivity is discretized over a grid of length n_h using importance sampling. This allows the model to replicate observed separation rate profiles across labor income deciles by projecting the data rates onto the productivity grid. Idiosyncratic transition rates satisfy $\mathbb{E}\ \Delta f_h = 0$ and $\mathbb{E}\ \iota_h = \bar{\iota}$

[12] The ownership state is a parsimonious device to match top-end wealth concentration; entry and exit probabilities do not respond directly to taxes. Profit income $\Pi(h)$ is taxed at the flat rate $\tau^{\text{profit}} = 21.1\%$, the effective business income taxation for Mexico estimated by Bachas et al. (2023), with the revenue accruing to income-tax collections.

formal employment is :

$$
\begin{aligned}
V^s(a,h) = \max_{a',\ell} \Big\{ u(c, 1-\ell) + \beta \mathbb{E}_h \Big[f_h V^f(a',h') + (1-f_h) V^s(a',h') \Big] \Big\}, \\
\text{s.t.} \qquad (1+\tau^c)c + a' = \underbrace{\Big[1 + (1-\tau^k)r\Big]}_{\equiv \tilde{R}} a + \ell w^s h^s + \tau^T, \\
\ln h' = \rho \ln h + \sigma_h \epsilon'_h, \quad \epsilon'_h \overset{\text{iid}}{\sim} \mathcal{N}(\mu_h, 1), \\
c \geq 0, \quad a' \geq 0, \quad \ell \in [0,1].
\end{aligned}
\tag{15}
$$

The expectation $\mathbb{E}_h$ is conditional on current productivity h, and $\tilde{R} \equiv 1 + (1-\tau^k)r$ is the gross after-tax return on assets. The condition $a' \geq 0$ imposes the borrowing constraint.

Let u_c denote marginal utility of consumption and u_2 marginal utility of leisure, the second argument of $u(c, 1-\ell)$. At an interior solution, the informal labor supply satisfies $\frac{w^s h^s}{1+\tau^c} \cdot u_c = u_2$.

4.4.2 Formally Employed Households' Problem

Households with a formal job can work in both sectors. They choose total hours ℓ and the formal share κ. Their allocation reflects productivity differences across sectors and the progressive tax on formal earnings. Their problem is

$$
\begin{aligned}
V^f(a,h) = \max_{a',\ell,\kappa} \Big\{ u(c, 1-\ell) + \beta \mathbb{E}_h \Big[\iota_h V^s(a',h') + (1-\iota_h) V^f(a',h') \Big] \Big\}, \\
\text{s.t.} \qquad (1+\tau^c)c + a' = \tilde{R} a + y^{\text{net}}(\kappa, \ell, h) + \tau^T + (1 - \tau^{\text{profit}})\,\Pi(h), \\
y^{\text{net}}(\kappa, \ell, h) = (1-\kappa)\ell w^s h^s + \tau^L \bar{y}^{\tau^P} \Big(\kappa \ell w^f h^f \Big)^{1-\tau^P}, \\
\ln h' = \rho \ln h + \sigma_h \epsilon'_h, \quad \epsilon'_h \overset{\text{iid}}{\sim} \mathcal{N}(\mu_h, 1), \\
c \geq 0, \quad a' \geq 0, \quad (\ell, \kappa) \in [0,1]^2.
\end{aligned}
\tag{16}
$$

Here $y^{\text{net}}(\kappa, \ell, h)$ is after-tax labor income from formal and informal work. Profit income $\Pi(h)$ is positive only for households in the ownership state. At an interior solution, total labor supply and the allocation of labor across sectors satisfy

$$
\begin{aligned}
u_2 &= \frac{(1-\kappa) w^s h^s + \tau^L \bar{y}^{\tau^P} (1-\tau^P) \ell^{-\tau^P} (\kappa w^f h^f)^{1-\tau^P}}{1+\tau^c} \cdot u_c, \\
\ell w^s h^s &= \tau^L \bar{y}^{\tau^P} (1-\tau^P) \kappa^{-\tau^P} (w^f \ell h^f)^{1-\tau^P}.
\end{aligned}
\tag{17}
$$

The first condition governs total hours. The second equates the marginal after-tax return to

formal work with the return to informal work. The inclusion and evasion margins of the static model therefore remain present even when employment status does not change. Taxes can change the formal share of hours among households that already have access to formal employment.

4.5 Formal Labor Agency

The quantitative model also allows policy to affect access to formal employment through vacancy creation. A representative labor agency posts vacancies according to the surplus generated by formal labor. Changes in that surplus therefore affect labor market tightness and the rate at which workers enter formal employment.

The agency supplies formal intermediate goods and is owned by households in the ownership state, who receive its net surplus. Let $L^f = \int \kappa \ell h \, d\mu(a, h, F)$ denote aggregate formal labor in efficiency units, where $\ell^f \equiv \kappa \ell h$ is individual formal labor in efficiency units and $\mu(a, h, F)$ is the distribution of households over assets, productivity, and employment status. The agency's recursive problem is:

$$\begin{aligned} J(L^f) = \max_{V} \; & -c_v V + (\Omega p^f - w^f) L^f + \tilde{R}^{-1} J(L^{f'}) \\ \text{s.t.} \quad & L^{f'} = \int \left[(1 - \iota_h) \, \mathbf{1}_{\{F=1\}} + f_h \mathbf{1}_{\{F=0\}} \right] \ell^{f'} \, d\mu(a, h, F), \end{aligned} \tag{18}$$

where $J(L^f)$ is the value of the labor agency, and c_v is the cost of posting a vacancy. The agency discounts future payoffs at the gross after-tax asset return $\tilde{R}$. In a stationary labor market, $L^f = L^{f'}$, and the zero-profit condition is

$$c_v = \tilde{R}^{-1} \frac{\left(\Omega p^f - w^f \right) L^f}{V} \tag{19}$$

The cost of posting one vacancy therefore equals the expected average revenue generated by a formal hire. Vacancy posting changes labor market tightness and thereby the stationary distribution of households with formal employment.

The labor agency receives p^f per unit of formal output and pays w^f per efficiency unit of formal labor. Search frictions leave a range of wages between households' and the agency's reservation values. Under Nash bargaining, household-specific outside options would make wages depend on

both wealth and productivity. We instead use the wage rule

$$w^f = w^s + \eta \left(\Omega p^f - w^s \right). \tag{20}$$

where η is the share of the flow surplus from a formal match accruing to the worker.[13]

Households' labor choices affect the surplus from formal employment and therefore vacancy posting and job finding. Optimal progressivity changes little when η varies by ± 0.15 around its baseline value of 0.60. The optimum shifts by $+0.01$ and -0.005 respectively, as reported in Appendix D.5.

4.6 Equilibrium

A stationary equilibrium combines optimal household choices, competitive factor prices, vacancy posting, and an invariant distribution $\mu(a, h, F)$. Household asset holdings finance the capital stock, and labor supplied in each sector equals labor demand. Let g^a, g^ℓ, g^κ, and g^c denote the household policy function for next period's assets, total hours, the formal share of hours, and consumption. The government balances its budget:

$$\begin{aligned} G + \tau^T = \int \Big\{ & T\left(g^\kappa(a,h,F) g^\ell(a,h,F) w^f h^f \right) + \tau^c g^c(a,h,F) \\ & + \tau^k r \cdot a + \tau^{\text{profit}} \Pi(h) \Big\} \, d\mu(a,h,F). \end{aligned} \tag{21}$$

In the benchmark and the main policy experiments, government consumption is fixed at $G/Y = 15\%$ and the lump-sum instrument τ^T adjusts to balance the budget. Appendix G.2 gives the full equilibrium definition and market-clearing conditions; Appendix D.2 considers alternative budget rules.

4.7 Calibration

We calibrate the model to Mexican data on sectoral earnings, employment transitions, wealth and consumption inequality, labor market outcomes, and public finances. The empirical analysis provides direct information on the two margins at the center of the model. Employment transitions describe access to formal jobs, while sectoral earnings help determine the relative payoff to informal work. Existing evidence does not directly identify the formal and informal labor supply responses needed for the policy experiments.

[13] Similar wage rules are used in Gornemann et al. (2016), Den Haan et al. (2018), and Graves (2025).

Households have additively separable preferences over consumption and leisure with constant relative risk aversion. Following King et al. (1988), these preferences allow wealth to affect labor supply. Flow utility is

$$u(c, 1-\ell) = \frac{c^{1-\zeta} - 1}{1-\zeta} + \Phi \frac{(1-\ell)^{1-\phi} - 1}{1-\phi}. \tag{22}$$

We use the simulated method of moments to match aggregate and cross-sectional evidence for Mexico in 2024. Table 2 reports the targeted moments and the parameters they help discipline. Table 3 reports the remaining parameters. The parameter and moment pairings summarize the joint calibration rather than one-to-one identification, since each parameter can affect several moments through household choices and equilibrium prices.

Inequality and Redistribution. The wealth targets help detetermine saving behavior and wealth inequality. The capital-to-output ratio disciplines the discount factor β. Following Castaneda et al. (2003), the stochastic ownership state helps reproduce wealth concentration at the top. Only top-productivity formal workers can enter this state. We calibrate the entry rate π^{up} so that owners account for 0.62% of households, matching the share of formal employers with six or more workers among the employed in the ENOE (2024). The wealth Gini disciplines σ_{LR}, the long-run standard-deviation of log efficiency units. This persistent heterogeneity generates differences in earnings and saving through the rest of the distribution.

Sectoral earnings moments discipline the relative payoff to informal work and how workers sort across sectors. We target the formal–informal gap in mean log earnings, the persistent component of this gap in log-earnings variance, and the ratio of informal to after-tax formal hourly wages. Together, these moments help determine the level and curvature of informal returns.

The joint calibration yield $\varkappa = 1.87$, $\Xi = 0.81$ and $\xi = 0.92$. The formal productivity scale fixes the units of measurement. We set $\Omega = 2.12$, so average gross formal earnings per full-time-equivalent formal worker equal one in the benchmark and the tax schedule takes its familiar level form. Before accounting for lower returns to productivity in informal work, the informal technology is about 12% less productive than the formal one, with $\varkappa/\Omega = 0.88$.

The model closely matches the targeted earnings gaps. Mean log earnings are 0.72 higher among formal workers in the data and 0.71 higher in the model. The persistent component of log-earnings variance is 0.17 *higher* among informal workers in both the data and the model. We

Table 2: Calibration targets and parameter values

Target	**Data**	**Model**	**Parameter**	**Description**	**Value**
K/Y	2.44	2.46	β	discount factor	0.92
Gini wealth	0.72	0.76	σ_{LR}	long-run s.d. of log efficiency units	1.00
owner share of households (%)	0.62	0.62	π^{up}	ownership entry	0.022
mean log-earnings gap	0.72	0.71	$\varkappa$	informal linear productivity	1.87
log-earnings variance gap, persistent	0.17	0.17	ξ	informal skills curvature	0.92
informal/formal net wage ratio	0.92	0.90	Ξ	informal skills scaling	0.81
Gini consumption	0.41	0.41	τ^P	earnings tax progressivity	0.015
L (hours / time endowment)	0.414	0.414	Φ	leisure weight	0.33
informality rate (headcount)	0.544	0.564	c_v	vacancy cost	0.015
avg. job-finding rate	0.159	0.168	χ	matching function TFP	0.159
separation rates	see Fig. 6a		ι_h	h-specific sep. rates	see[a]
job-finding rates	see Fig. 6b		Δf_h	h-specific job-f. rate diffs	see[a]
consumption tax revenues / Y	0.056	0.057	τ^c	consumption tax rate	0.083
income tax revenues / Y	0.074	0.073	τ^k	capital tax rate	0.081
tax rate at mean formal earnings	0.128	0.125	τ^L	labor-tax scale	0.875

Notes: Baseline calibration for Mexico. Parameter–moment pairings summarize the joint calibration. K is capital and Y is output. The owner share is the share of formal employers with six or more workers among the employed. The mean log-earnings gap is formal minus informal; the variance gap is the persistent component of the informal-minus-formal gap in the variance of log earnings (Section 2.3). The wage ratio is the informal to after-tax formal hourly wage at mean formal earnings, from the salaried lag-matched joint estimate of Section 2.3. L is average weekly hours worked over a 100-hour endowment. Informality is the share of employed workers in informal jobs; in the model, a worker is informal if less than half of her hours are formal. The ownership entry probability is the effective per-period probability. Income-tax revenue includes labor, capital, and profit taxes. The last row reports the average tax rate at mean formal earnings. Labor-market targets use the National Survey of Occupation and Employment (ENOE, 2024); fiscal targets use OECD Taxing Wages (OECD, 2026) and OECD Revenue Statistics (OECD, 2025). The capital-output ratio is the 2023 net capital stock per unit of GDP from the European Commission's AMECO database; the wealth Gini is from UBS (2025); the consumption Gini is from ENIGH 2024.

[a] Transition rates are annual flows between informal and formal employment by labor-income decile, assigned across productivity types using the profiles in Figure 6.

target the persistent component because the baseline income process does not include separate transitory shocks. The larger dispersion of informal earnings arises from sorting and composition. The common persistent productivity process affects both sectors, while concave informal returns, $\xi < 1$, compress rather than amplify productivity differences.[14]

The calibrated value $\xi = 0.92$ is somewhat above the direct panel estimate of about 0.8. In the baseline, ξ is estimated jointly with the other productivity parameters to match the sectoral earnings distribution. Appendix C instead fixes it at the panel estimate and evaluates the resulting policy implications.

We choose tax progressivity τ^P to match consumption inequality, conditional on the model's wealth and earnings distributions. The tax schedule determines how earnings risks passes through to disposable income and consumption. Panel 6d later compares the resulting schedule with Mexican fiscal data that are not targeted in the calibration.

Labor Market Outcomes. The labor market targets capture how much households work, where they work, and how they move between sectors. Average hours in the ENOE (2024) are 41.4 per week, or 0.414 of a 100-hour weekly endowment. Given the curvature of leisure utility, the weight Φ matches this average exactly.

We also target the informality rate, the average rate of finding a formal job, and job finding and separation rates across labor income deciles. Informal workers account for 0.544 of employed workers in the data. In the model, a worker is classified as informal when less than half of her hours are formal. Annual transitions are measured among workers employed at two ENOE interviews one year apart, from 2023 to 2024.. On average, 15.9% of informal workers enter formal employment and 16.9% of formal workers move to informality within a year. These sector shares and transition rates help determine the vacancy costs c_v, matching efficiency χ, and the productivity-specific transition schedules Δf_h and ι_h. Together with average hours, they discipline labor supply and access to formal employment.

Fiscal Tools. We calibrate the labor tax scale τ^L in (8) using the observed average tax rate of 12.77% at mean formal earnings, consistent with OECD (2026). The consumption tax τ^c targets consumption tax revenue, and the capital income tax τ^k targets income tax revenue. Both revenue

[14] Appendix C reports two robustness calibrations. One introduces the sector-specific transitory earnings risk measured in the panel and targets the full variance gap. The other fixes ξ at its panel estimate together with the panel's earnings persistence. In both cases, welfare-maximizing progressivity remains several times its benchmark level.

targets are shares of GDP averaged over 2021–24 from the OECD Revenue Statistics (OECD, 2025). Progressivity τ^P targets the consumption Gini, while the lump-sum instrument τ^T balances the government budget.

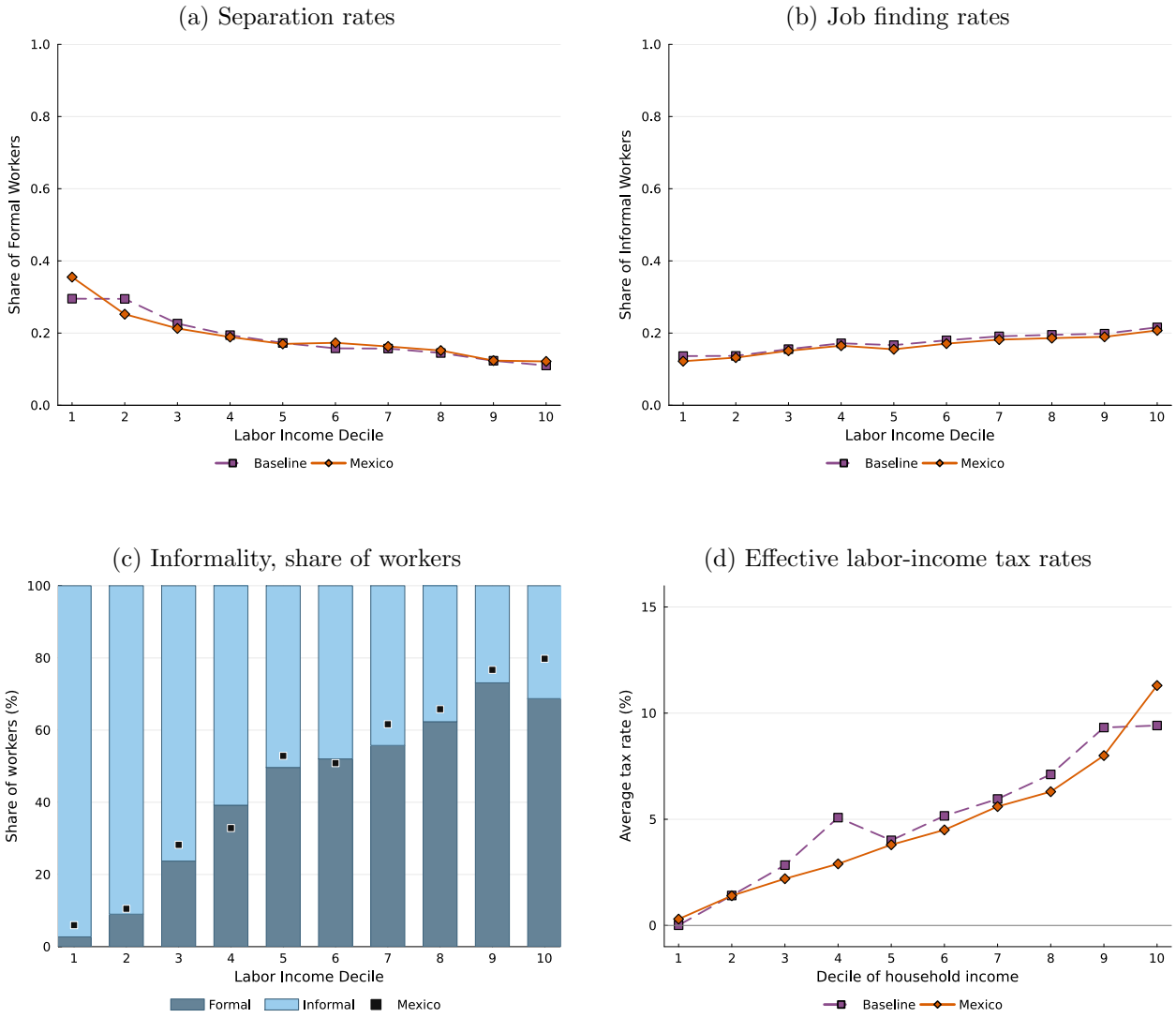
Government expenditure is set at 15% of GDP. Consumption taxes provide 5.7 percentage points and taxes on household income, including labor, capital, and business profits, provide 7.3 percentage points. A uniform lump-sum levy finances the remaining share of about 2 percent of GDP. This instrument stands in for Mexico's substantial non-tax financing. Total public expenditure is around 25% of GDP, with roughly 10 percentage points financed through oil revenues, other non-tax revenues, and fiscal deficits. In the policy experiments, the lump-sum instrument rebates or collects changes in revenue uniformly. Its level affects household resources and hence labor supply and saving, even though it creates no direct marginal tax wedge.

Remaining Parameters. Table 3 reports the remaining parameters. Relative risk aversion is $\zeta = 2$. The leisure curvature ϕ is set so that the Frisch elasticity at average hours is 0.3, following Leyva and Urrutia (2020). The capital exponent is $\alpha = 0.23$, annual capital depreciation is $\delta = 0.06$, and the elasticity of substitution between formal and informal inputs is $\varepsilon = 7.70$. The matching elasticity is $\psi = 0.6$, and the wage rule parameter is $\eta = 0.6$. The span of control parameter is set to $\nu = 0.96$.

The baseline income process has annual persistence $\rho = 0.98$. Persistence affects wealth and consumption through saving and self-insurance but is not identified by the stationary productivity distribution alone. Appendix C shows that optimal progressivity remains several times its benchmark level when we introduce the sector-specific transitory risk estimated from the panel or impose the panel estimate of ξ together with lower earnings persistence.

4.7.1 Fit of the Baseline Economy

Figure 6 summarizes the dimensions of the calibration most closely related to the policy mechanism. The separation and job finding profiles are calibration targets. Low-income workers face higher separation rates and lower rates of entry into formal employment, making informal work a more important outside option. The informality profile combines these differences in access with households' allocation of labor across sectors. The final panel provides an untargeted comparison between the effective tax rates implied by the model and Mexican fiscal data from the Secretaría de Hacienda y Crédito Público (SHCP).



Notes: In Panels 6a–6c, data are from Mexico's National Survey of Occupation and Employment (ENOE, 2024), and workers are ranked by labor income. Panels 6a and 6b show transition profiles targeted in the baseline calibration: annual flows between informal and formal employment by labor-income decile. Panel 6c shows the shares of formal and informal workers in the model (bars) and the formal share in the data (squares); in the model, a worker is informal if less than half of her hours are formal. Panel 6d is not targeted. It shows the model's labor-income tax paid over total gross household income (labor and capital income), by decile of total gross income, excluding entrepreneurs, and the income tax on salaries over total gross household income, by decile of households, as estimated by the Secretaría de Hacienda y Crédito Público (SHCP) from ENIGH 2024 in the scenario that assigns the tax only to earners affiliated with social security (*Distribución del pago de impuestos y recepción del gasto público por deciles de hogares y personas*, 2026). Appendix D.7 reports the version that includes entrepreneurs and the tax on their profits.

Figure 6: Baseline economy labor-market gradients and effective tax rates

Table 3: Remaining parameters for the baseline economy

Parameter	Symbol	Value	Parameter	Symbol	Value
Risk aversion	ζ	2.00	Income persistence	ρ	0.98
Frisch elasticity	$\frac{1-\ell}{\ell}\phi^{-1}$	0.30	Capital depreciation	δ	0.06
Formal wage rule	η	0.60	Matching function exp.	ψ	0.60
Span-of-control (DRS)	ν	0.96	Capital exponent in composite	α	0.23
Profit tax rate	τ^{profit}	0.211	Fall-distribution rate	π^{unlucky}	2.00
Formal productivity (normalization)	Ω	2.12	Ownership exit rate	π^{fall}	0.21
Formal–informal elast. of substitution	ε	7.70			

Notes: Depreciation and income persistence are annual. Preferences follow Leyva and Urrutia (2020); the Frisch elasticity is evaluated at average hours, $\ell = 0.414$, which gives $\phi = 4.72$. π^{up} (Table 2) and π^{fall} govern entry and exit; π^{unlucky} is the exponential rate governing productivity after exit. The exit probability is the effective per-period probability; the entry probability is calibrated to the owner share of households. Formal productivity Ω normalizes average gross formal earnings per full-time-equivalent formal worker to one at the benchmark.

Formality rises across income deciles in both the data and the model, whether measured by the share of workers in Panel 6c or the share of hours in Appendix Panel D.4a. At the same time, substantial informal work remains at the top of the distribution. This pattern does not arise only from the calibrated differences in access to fomal jobs. In the economy with common job finding and separation rates repoted in Appendix D.6, informality still declines with income. In the aggregate, the informal share of hours is 0.581 in the model and 0.514 in the data. Within the informal sector, the ratio of the 20th to the 80th percentile of earnings is 0.20 in the model and 0.30 in the data.

Panel 6d provides an untargeted check on the benchmark tax schedule. Progressivity is calibrated to consumption inequality rather than directly to observed effective tax rates. Nevertheless, the model generates effective labor income tax rates across income deciles that are close to those estimated by Mexico's Ministry of Finance from ENIGH 2024. This provides an external check on the degree of progressivity generated by the benchmark calibration.

5 Optimal Progressivity with Informality: Quantitative Results

The static model shows how the welfare effects of progressivity are shaped by households' ability to move labor between formal and informal activities. Subsidies to low formal earnings create an inclusion margin, while higher marginal rates eventually induce high earners to shift labor outside the tax base. The quantitative model allows these responses to operate through both labor allocation and access to formal employment, while households self-insure and prices, vacancy creation, and

the government budget adjust in equilibrium.

We use the calibrated model to ask how these informal-sector responses shape the effects of progressivity in Mexico. We first revisit the experiment of the static model and examine how inclusion and evasion affect the formal tax base. We then compare progressivity with changes in labor and capital tax levels, study how the balance between inclusion and evasion changes with earnings inequality, and assess its sensitivity to the earnings dynamics and informal returns measured in the panel.

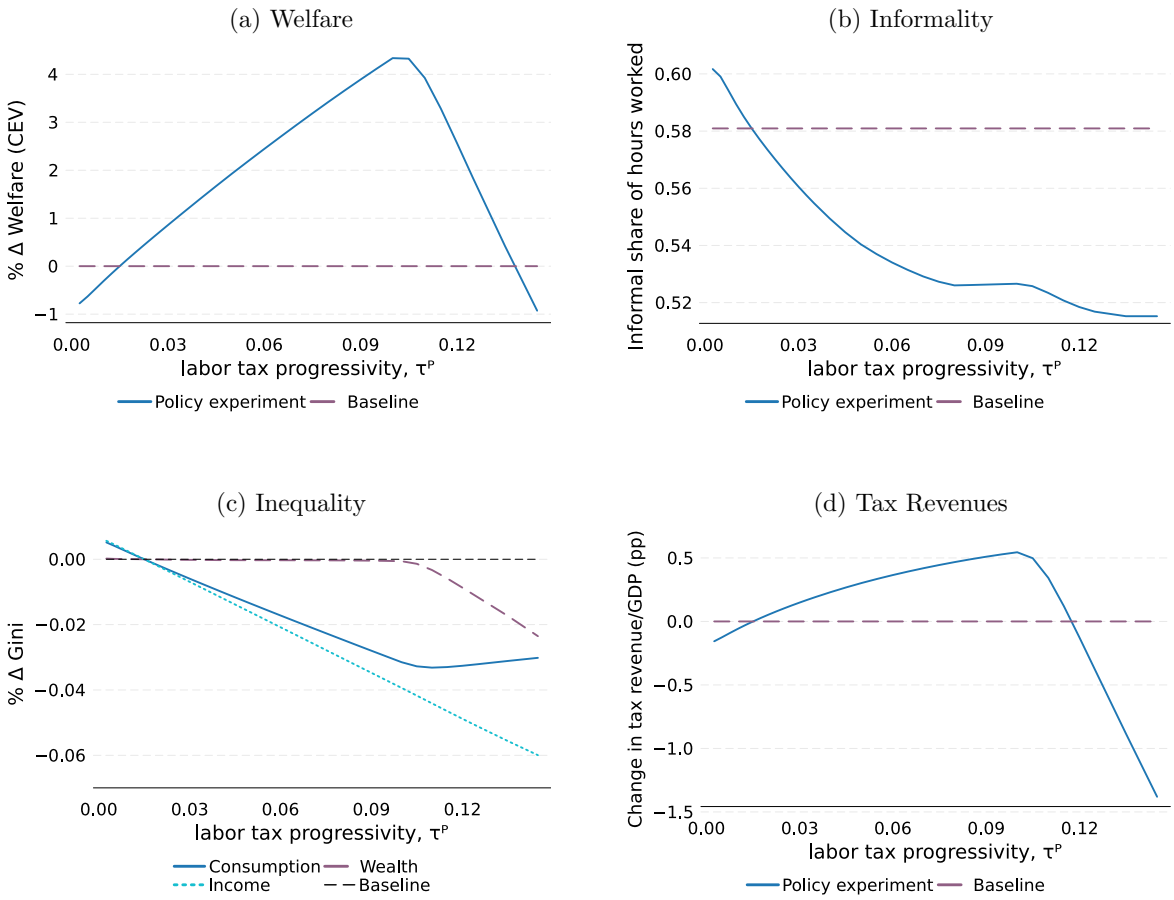
In the benchmark experiments, government consumption remains at its calibrated share of output, $G/Y = 15\%$. Every reform balances the government budget: the lump-sum instrument returns additional revenue net of spending to households or collects any shortfall. We maximize utilitarian welfare over each instrument's explored range, varying one tax parameter at a time and holding other tax and structural parameters fixed. Prices, household choices, and the stationary distribution adjust to each reform. Welfare includes consumption and leisure and is evaluated over each economy's stationary distribution. Consumption-equivalent variation (CEV) measures the welfare change relative to the baseline. Appendix D.4 states the criterion.

5.1 Optimal Progressive Labor Taxation

Figure 7 revisits the main experiment of the static model in the calibrated economy. The static model produced a hump-shaped welfare response as the inclucion margin eventually gave way to evasion. The quantitative model preserves this result. Welfare peaks at $\tau^P = 0.10$, about six-and-a-half times the benchmark level of 0.015. At this point, welfare rises by 4.3% of consumption, the informal share of hours falls by 5.4 percentage points, consumption inequality falls by 3.2%, and tax revenue increases by about half a percentage point of GDP.

The welfare-maximizing degree of progressivity also maximizes revenue relative to GDP. Up to $\tau^P = 0.10$, welfare and revenue rise together, while aggregate informality declines. Beyond this point, welfare and revenue fall, indicating that further progressivity begins to erode the formal tax base. Informality behaves differently. It falls sharply up to $\tau^P \approx 0.08$, changes little around the welfare optimum, and continues to decline slowly beyond it. The welfare optimum therefore does not coincide with the minimum level of informality.

The allocation of formal labor in Panel 8a explains why aggregate informality and welfare behave differently beyond the optimum. Among households with formal access, the response closely resembles the mechanism in the static model. Households in the lower part of the productivity dis-

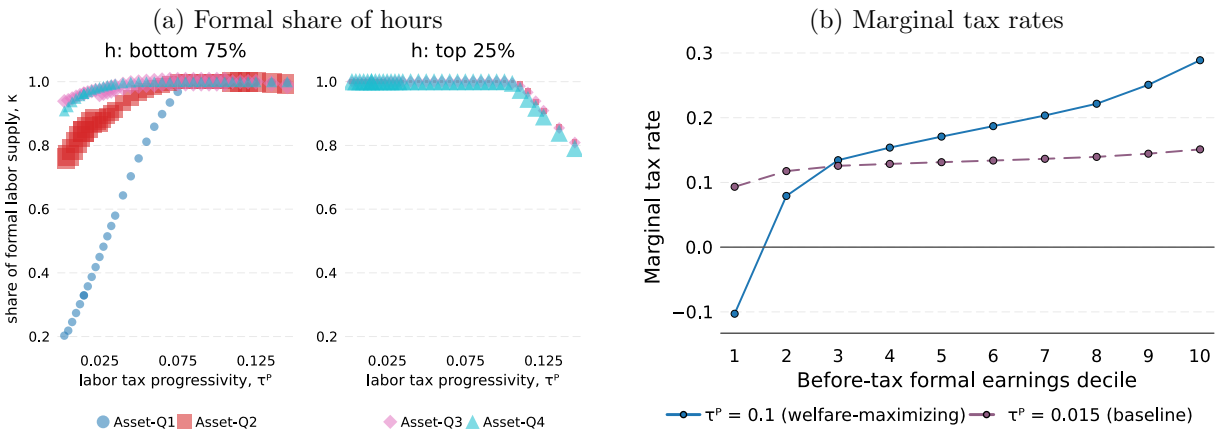


Notes: Labor-tax progressivity τ^P varies from its baseline of 0.015, with other tax and structural parameters fixed. Government consumption remains 15% of output, and the lump-sum transfer balances the budget. Welfare gains are steady-state consumption-equivalent variations (CEV, percent), including leisure, relative to the baseline. Informality is shown in levels. Gini changes are proportional changes from the baseline: −0.01 denotes a 1% decline. Panel (d) reports changes in tax revenues relative to GDP.

Figure 7: Optimal progressivity: welfare, informality, inequality, and revenue

tribution increase their formal share of hours as progressivity rises. The response is strongest among households with fewer assets, which initially allocate more labor to informal work. As subsidies to low formal earnings become more generous, these differences narrow and formal hours approach one across asset groups. This response accounts for the sharp decline in aggregate informality in Panel 7b at lower levels of progressivity.

Households in the top productivity quartile behave differently. Their formal share remains close to one across asset groups up to the welfare optimum, but begins to fall as progressivity rises further. The inclusion margin therefore dominates initially, while the evasion margin becomes relevant around the optimum.



Notes: Government consumption remains 15% of output, and the lump-sum transfer balances the budget as progressivity varies; other tax parameters are fixed. Panel (a) shows the formal share of hours κ of households with a formal job by asset quartile, separately for the bottom 75% and top 25% of productivity, entrepreneurs included in the top group. Marker area is proportional to the cell's share of all households. Panel (b) compares the baseline and welfare-maximizing marginal tax schedules at mean gross formal earnings within each decile of the baseline distribution of formal earners.

Figure 8: The inclusion and evasion margins

The quantitative model adds another response that is absent from the static framework. As formal hours fall among high-productivity households, the surplus earned by the labor agency per unit of formal labor rises and vacancy posting increases. Job finding therefore improves for households without formal access. Those who obtain a formal job supply most of their hours formally.The expansion in formal access more than offsets the reallocation toward informal work among high-productivity households, which explains why aggregate informality in Panel 7b continues to fall even after welfare and revenue begin to decline. Appendix D.1 reports these labor-market responses.

This distinction is important for interpreting the aggregate informality rate. Beyond the optimum, lower informality no longer implies that the distortion created by progressivity is becoming smaller. The composition of informal activity changes as high-productivity labor begins to leave the tax base. What matters for welfare and revenue is therefore not only how much informal labor there is, but also which households supply it.

Panel 8b translates the reform into marginal tax rates along the distribution of gross formal earnings. The baseline schedule is nearly flat, with rates between 9% and 15% across deciles. The welfare-maximizing schedule has a negative marginal rate in the bottom decile, crosses the baseline between the second and third deciles, and reaches 29% in the top decile. Average taxes are negative for roughly the bottom sixth of formal earners, whose earnings are below a quarter of the mean.

Higher taxation at the top thus finances subsidies to formal work at the bottom.

Additionally, Panel 7c reveals a second divergence. At the optimum, labor-income and wealth Ginis are 3.9% and 0.06% below the benchmark, and both continue to fall as progressivity rises. Consumption inequality reaches its minimum just beyond the optimum and then increases slightly. At the same time, revenue in Panel 7d falls as evasion contracts the formal tax base and the lump-sum fiscal instrument collects more from households. Consumption inequality nevertheless remains below its benchmark throughout the explored range and is 3.0% lower at the top of the grid. Lower inequality, like lower aggregate informality, can therefore coexist with welfare losses once the tax base begins to erode.

Reallocation across sectors also affects measured TFP, which in the model depends on the informal share of labor measured in efficiency units (Appendix G.1). The effect is small at the welfare optimum, where TFP is 0.004% above its benchmark value. At the top of the progressivity grid, $\tau^P = 0.145$, it is 0.26% lower. The difference reflects the productivity of the household reallocating labor. Inclusion moves lower-productivity labor toward formal production, while evasion moves high-productivity labor in the opposite direction. The welfare gains around the optimum therefore arise mainly through redistribution, insurance, and changes in the formal tax base rather than through large productivity effects.

The direction of these responses is consistent with evidence from policies that change the relative payoff to formal employment. In Mexico, a temporary wage subsidy tied to formal employment increased formal work partly by drawing workers out of informality (Abel et al., 2022). The opposite response appears when policy makes formal work relatively less atrractive. An income-tested transfer in Uruguay reduced formal employment (Bérgolo and Cruces, 2021), while Seguro Popular in Mexico reduced formal registration (Bosch and Campos-Vázquez, 2014).

Evidence on the evasion margin is more limited. Slonimczyk (2012) finds that Russia's 2001 tax reform reduced informal employment, with stronger responses among individuals receiving larger tax reductions. Using the same reform, Gorodnichenko et al. (2009) document substantial changes in tax evasion associated with voluntary compliance.

5.2 Progressivity Compared with Other Tax Instruments

The static model also suggests that the shape of labor taxation matters. Greater progressivity raises the return to formal work at low earnings, while a higher labor-tax level reduces it throughout the distribution. We now ask whether this distinction remains quantitatively important once the

Table 4: Fiscal reforms and the trade-off between welfare and informality

	Reform (pp)	Welfare CEV (%)	Informality change (pp)	Revenue/GDP change (pp)	CEV at equal revenue (%)
G/Y fixed, lump-sum transfers adjust					
Progressivity, τ^P	+8.5	4.34	-5.43	+0.54	4.34
Labor-tax level, $1-\tau^L$	+10.0	6.33	+19.65	+3.28	1.42
Capital tax, τ^k	+9.0†	1.62	-0.05	+0.77	1.16
Transfer fixed, government consumption adjusts					
Progressivity, τ^P	+10.0‡	2.65	-6.03	+0.11	2.65
Labor-tax level, $1-\tau^L$	–	–	–	–	-0.11
Capital tax, τ^k	–	–	–	–	-0.18

Notes: Each row maximizes welfare over the explored range for one instrument, holding other tax parameters fixed. Changes are relative to the baseline. CEV denotes steady-state consumption-equivalent variation, including leisure. Informality is the informal share of hours worked. Government consumption has no direct utility value. The labor-tax level $1-\tau^L$ is the average tax rate at mean formal earnings. † marks a grid boundary, not an interior optimum. In the second block, welfare is maximized subject to government consumption remaining at least at its baseline share of output, 15%; ‡ marks an optimum constrained by this spending requirement. – marks no reform: the best reform is the baseline itself. The last column reports each instrument's CEV at the reform that changes revenue/GDP by as much as the progressivity reform in the same block (+0.54 and +0.11 percentage points). The grids are $\tau^P \in [0.0025, 0.145]$, $\tau^L \in [0.755, 0.885]$, and $\tau^k \in [0.071, 0.171]$. Appendix D.3 also reports fixed absolute government spending.

reforms raise revenue and the economy adjusts in equilibrium. We compare progressivity with changes in labor and capital tax levels under the same government budget rule, holding G/Y fixed and returning changes in revenue net of spending to households.

Table 4 reports the best reform for each instrument over the explored ranges. Raising the labor-tax level ($1-\tau^L$) by 10 percentage points delivers a welfare gain of 6.3% of consumption. A 9-percentage-point increase in the capital income tax delivers a gain of 1.6%, with welfare still rising at the upper edge of the grid. Greater progressivity yields 4.3%. These reforms, however, have sharply different effects on informality. It falls by more than five percentage points under progressivity, rises by almost twenty points under the higher labor-tax level, and changes little under capital taxation.

The larger welfare gain from the labor-teax level reflects the much larger amount of revenue it raises and returns to households. Revenue increases by 3.3 percentage points of GDP, about six times the increase at the progressivity optimum. The last column of Table 4 therefore compares the

instruments at equal revenue. For a common revenue increase of 0.54 percentage points of GDP, the welfare gain is 1.4% under the higher labor-tax level and 1.2% under the higher capital tax. Progressivity yields 4.3%, three to four times as much. The difference is therefore not simply the amount of revenue redistributed.

Progressivity is also the only instrument whose best reform does not reduce measured TFP. TFP is 0.004% above its benchmark at the progressivity optimum, 0.48% below it at the best labor-tax-level reform, and essentially unchanged at the upper edge of the capital-tax grid.

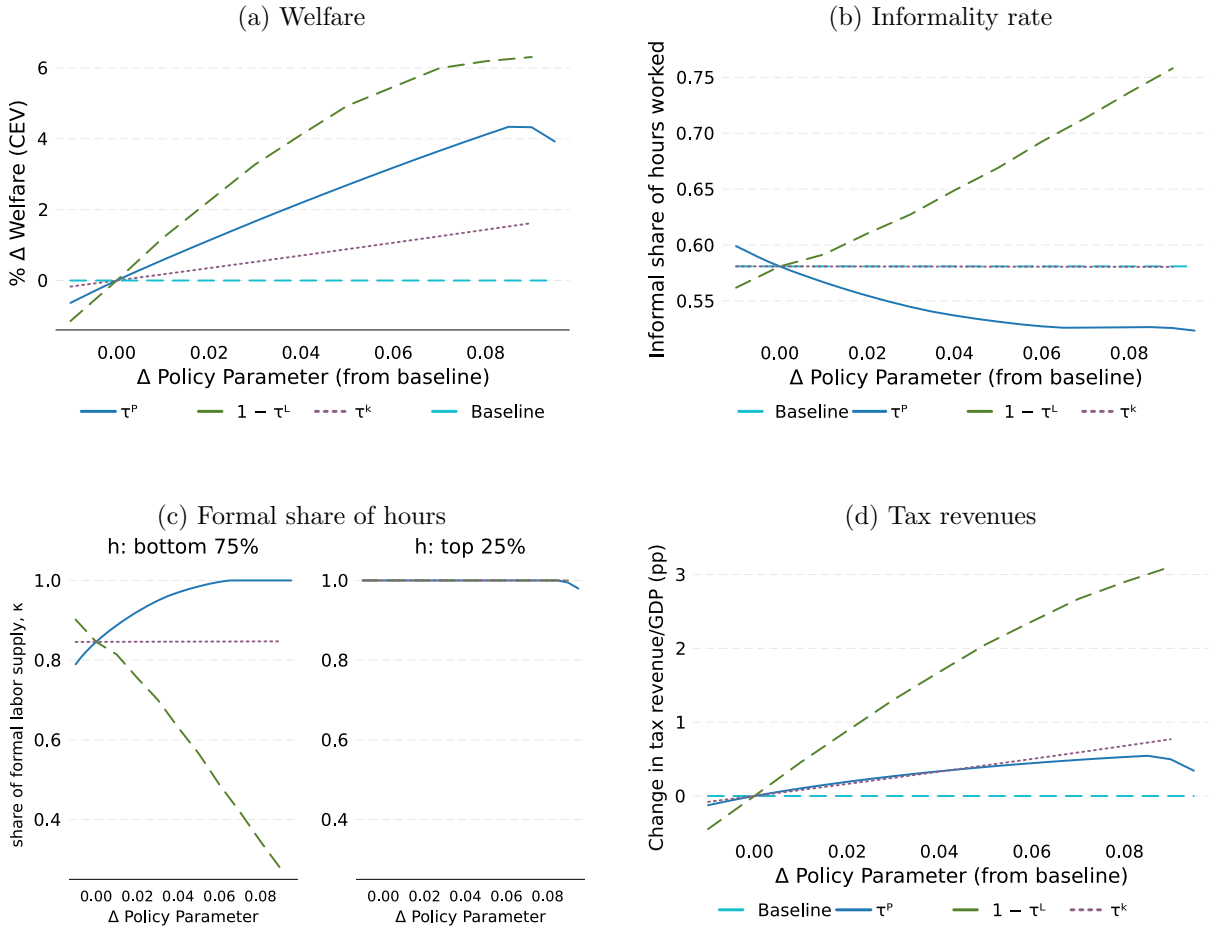
Figure 9 shows the labor responses behind these differences. Panel 9c shows that progressivity raises formal labor among households in the bottom three quarters of the productivity distribution, while the top quarter changes little until the evasion margin becomes active. A higher labor-tax level produces the opposite response among lower-productivity households because it reduces the return to formal work at all earning levels. Households in the top productivity group remain almost fully formal over the explored range. As a result, informality in Panel 9b falls under progressivity but rises throughout the labor-tax-level experiment. Capital taxation produces almost no change in sectoral labor allocation. Panel 9d shows the corresponding revenue responses.

The second block of Table 4 asks whether the gains from progressivity depend on returning additional revenue to households. We hold transfers fixed and allow government consumption to absorb changes in revenue. Because government consumption has no direct utility value in the model, we require it to remain at least at its baseline share of output. Otherwise welfare could rise simply because government consumption falls. Under this closure, progressivity continues to improve welfare. The best admissible reform has $\tau^P = 0.115$, raises welfare by 2.65%, and reduces informality by 6.0 percentage points. Raising either the labor-tax level or the capital tax no longer improves welfare. Only reductions in these tax rates generate welfare gains, and these are ruled out by the spending floor. At the revenue increase generated by the progressivity reform, higher labor and capital taxes reduce welfare by 0.11% and 0.18%, respectively.

The distinction therefore survives when additional revenue is not returned to households. What separates the instruments is not only how much they redistribute, but how they change the formal tax base.

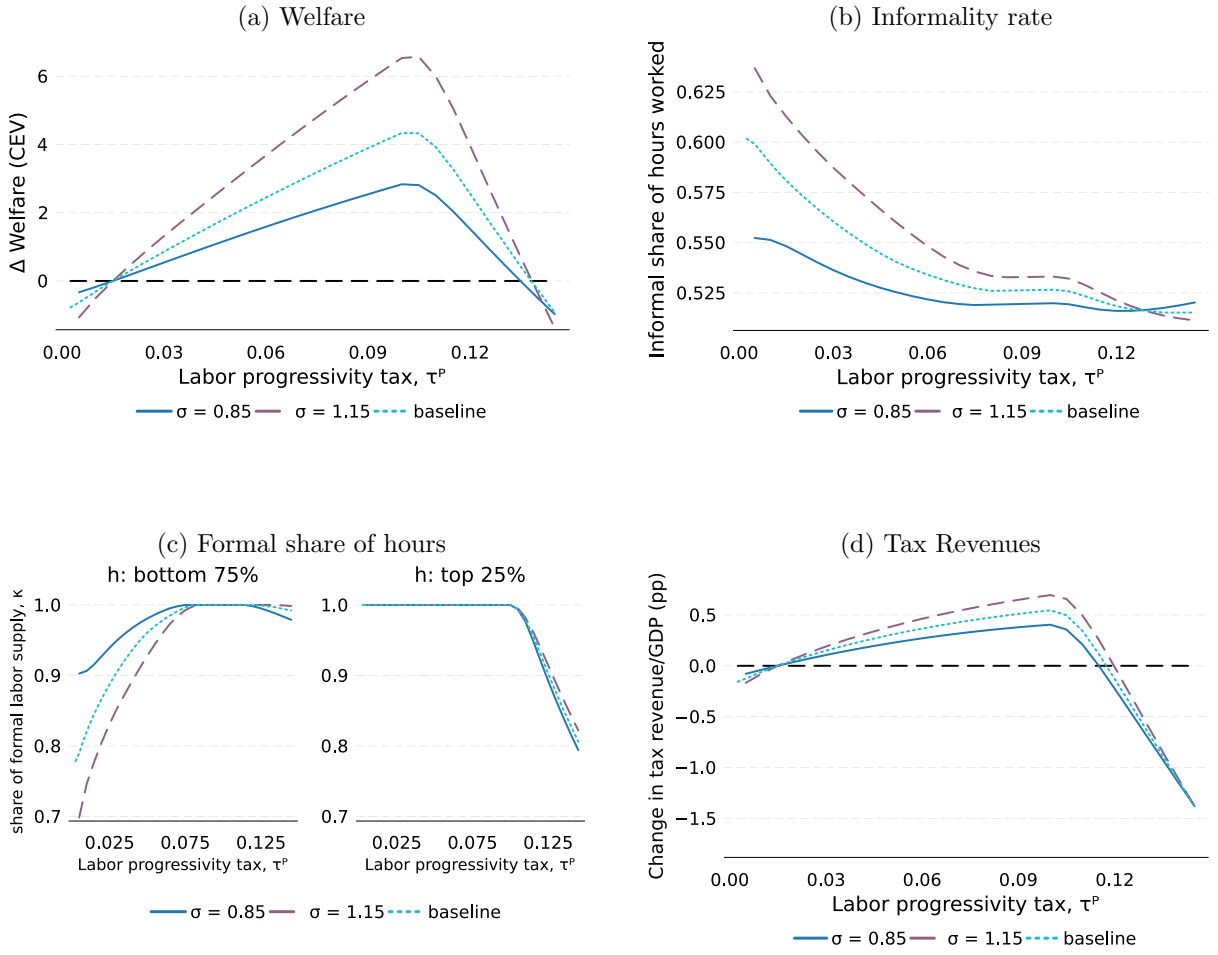
5.3 Inequality and Optimal Progressivity

Section 3 showed that greater productivity dispersion can raise informality by placing more households where informal work is relatively attractive. The quantitative model allows us to ask how the



Notes: Each tax instrument varies separately, with other tax and structural parameters fixed. The labor-tax level $1 - \tau^L$ is the average tax rate at mean formal earnings. A horizontal-axis change of 0.01 is one percentage point. Government consumption remains 15% of output, and the lump-sum transfer balances the budget. Welfare gains are steady-state consumption-equivalent variations (CEV, percent), including leisure, relative to the baseline. Informality (the informal share of hours worked) and the formal-hours share κ are levels; panel (c) splits households by productivity. Panel (d) reports changes in tax revenues relative to GDP. The labor-tax-level profiles are shown up to +9 percentage points; Table 4 reports the interior welfare maximum of this instrument, at +10.

Figure 9: Progressivity compared with labor- and capital-tax levels



Notes: The economies differ in the long-run standard deviation of log productivity, $\sigma_{LR} = 0.85$, 1.00 (baseline), and 1.15; other structural parameters are fixed. Each economy's labor-tax schedule is specified relative to its own average gross formal earnings at baseline progressivity, held fixed as τ^P varies. Government consumption remains 15% of output, and the lump-sum transfer balances the budget. Welfare gains are steady-state consumption-equivalent variations (CEV, percent), including leisure, relative to each economy's own equilibrium at baseline progressivity $\tau^P = 0.015$. Informality (the informal share of hours worked) and the formal-hours share κ are levels; panel (c) splits households by productivity. Revenue changes are relative to each economy's own baseline.

Figure 10: Inequality and the welfare stakes of progressivity

same dispersion changes both sides of the response to progressivity.

The long-run standard deviation of log productivity is $\sigma_{LR} = \sigma_h/\sqrt{1-\rho^2}$. Figure 10 compares a low-dispersion economy with $\sigma_{LR} = 0.85$, the baseline with $\sigma_{LR} = 1.00$, and a high-dispersion economy with $\sigma_{LR} = 1.15$. All other structural parameters are held fixed. Each reform is evaluated relative to that economy's baseline tax system, and the labor-tax schedule is specified relative to its own average formal earnings.

These differences generate substantial variation in income and consumption inequality. Moving from the low- to the high-inequality economy raises the labor-income Gini from 0.37 to 0.49 and the variance of log consumption from 0.48 to 0.83. The wealth Gini instead falls from 0.79 to 0.73. Greater productivity risk raises precautionary saving, while equilibrium changes in interest rates and accumulation can compress the upper tail of wealth (Aiyagari, 1994; Benhabib et al., 2011; Huggett, 1996). The experiment therefore varies earnings risk and consumption inequality rather than every dimension of inequality simultaneously.

Panel 10a shows that the welfare-maximizing degree of progressivity changes very little across the three economies. The optimum is $\tau^P = 0.10$, 0.10, and 0.105 in the low-dispersion, baseline, and high-dispersion economies. The welfare stakes change much more. The gain from the optimal schedule rises from 2.8% of consumption to 4.3% and 6.6%, respectively. Welfare rises faster at low progressivity and falls more sharply beyond the peak when inequality is higher.

Greater dispersion strengthens both sides of the informality tradeoff. More households lie in the lower part of the productivity distribution, where formal jobs are less accessible and informal returns are relatively attractive. These households have more to gain from subsidies to formal earnings. At the same time, taxable income becomes more concentrated among high-productivity households, making the tax base more sensitive to their response to higher marginal rates.

Panel 10c shows the resulting changes in formal labor. With greater dispersion, the inclusion response is stronger at lower levels of progressivity, while the fall in formal labor at high progressivity is also more pronounced. Panel 10d shows the corresponding revenue response. Revenue rises more rapidly before the optimum and falls more sharply once evasion becomes important. Panel 10b reports the resulting aggregate informality profiles. These opposing responses make the welfare profile steeper while leaving its peak almost unchanged.

Changes in inequality at the optimum are similar across the three economies. Labor-income Ginis fall by 1.5 to 1.9 percentage points, while the variance of log consumption falls by 0.03 to 0.07, with larger declines in the more dispersed economies.

Greater earnings risk therefore increases the welfare consequences of progressivity much more than its welfare-maximizing level. The standard prescription is that higher uninsurable earnings risk calls for a more progressive schedule (Heathcote et al., 2017, 2020).

5.4 Robustness of the Policy Results

Table 5: Robustness of optimal progressivity

	Benchmark	G/Y fixed		G fixed	
Calibration	τ^P	Optimal τ^P	CEV (%)	Optimal τ^P	CEV (%)
Baseline	0.015	0.100	4.34	0.100	3.04
Fixed ξ	0.016	0.125	1.24	0.090	0.43
Transitory earnings risk	0.018	0.060	1.85	0.060	1.26

Notes: Each calibration is re-fitted before varying labor-tax progressivity τ^P. Welfare gains are steady-state consumption-equivalent variations (CEV), including leisure, relative to that calibration's benchmark. Lump-sum transfers balance the budget with government consumption fixed relative to output (G/Y) or in levels (G). The fixed-ξ calibration sets informal-return curvature at its panel estimate, $\xi = 0.80$, and annual earnings persistence at $\rho = 0.944$. The transitory-risk calibration matches sector-specific transitory earnings variances from Mexico's ENOE panel using a common shock, keeps the baseline persistence, and re-fits the dispersion of the persistent component. Appendix C reports calibration details.

We next examine whether these results survive alternative fiscal rules and calibrations of earnings risk and informal returns. Substantially higher progressivity remains welfare-improving under alternative fiscal rules and calibrations. Optimal progressivity is 0.10 under both the benchmark rule and fixed absolute spending, and 0.115 with fixed transfers and a floor on government consumption at its baseline share of output (Appendix D.2). Informality falls by 5.4 to 6.0 percentage points, with welfare gains of 4.34%, 3.04%, and 2.65%, respectively. With G/Y fixed, government consumption moves with output; with G fixed, it does not adjust. Without the spending floor, welfare under fixed transfers keeps rising with progressivity as government consumption shrinks. The preferred schedule is stable across these fiscal rules, while the gains depend on how the government uses revenue and how spending changes with output.

The alternative calibrations also favor progressivity several times above their benchmarks (Appendix C). Table 5 compares their preferred tax schedules and welfare gains under the benchmark and fixed-spending rules. Adding the estimated sector-specific transitory component yields an optimum of 0.06, compared with that calibration's benchmark of 0.018. The welfare gain falls to 1.85% under the benchmark budget rule. Setting both the curvature of informal returns and earnings persistence to their panel estimates yields an optimum of 0.125, compared with a benchmark of

0.016. The welfare gain is 1.2%. Under fixed absolute spending, that calibration's optimum is 0.09.

These calibrations help distinguish the insurance and labor-allocation channels. Households can use assets to absorb temporary earning losses, while persistent losses reduce resources over many periods and are harder to smooth through saving.. Matching the panel risk estimates therefore changes the wealth and consumption distributions as well as sectoral labor allocation. The transitory-risk calibration assumes a common shock with different sectoral loadings; the panel identifies the sectoral variances but not their correlation. Under the benchmark fiscal rule, both alternative calibrations favor substantially higher progressivity, with the transitory-risk result conditional on the common-shock assumption. The welfare gains and sectoral responses depend more strongly on the calibration. Informality falls by 7.2 percentage points at the optimum with ξ fixed, but rises by 0.3 points with transitory risk. In the latter calibration, almost every worker with a formal job already works only formal hours at the benchmark. This mutes the inclusion margin by leaving little informal work among these households to draw into formal employment (Appendix C.1).

Finally, Appendix D.5 varies the wage rule, labor-supply elasticity, and informal returns one at a time. Changing the wage-rule weight by ± 0.15 moves optimal progressivity by 0.005 to 0.01. The labor-supply elasticity matters more for the size of the gains. Halving the Frisch elasticity raises the optimum to 0.12 and halves the welfare gain. Doubling it lowers the optimum to 0.09 and raises the gain to 5.0%. Informal returns move the optimum most: lowering ξ from 0.92 to 0.87 raises it to about 0.135, while raising ξ to 0.97 lowers it to 0.07 and keeps about 70% of the baseline gain. When informal earnings rise more strongly with ability, high-productivity households can avoid taxes at lower cost. Informal returns therefore matter for tax design through their effect on evasion.

6 Conclusion

Progressive labor taxation can improve welfare and reduce informality. Negative income taxes draw low earners into the formal tax base, while higher marginal rates at the top eventually shift labor toward untaxed informal work. In our model calibrated to Mexico, the welfare-maximizing degree of progressivity is over six times the current level. The reform raises steady-state welfare by 4.3% of consumption and lowers the informal share of hours worked by 5.4 percentage points.

The comparison across fiscal instruments shows why the shape of taxation matters. Higher labor-tax levels redistribute through the revenue returned to households but increase informality.

Higher capital taxes redistribute with little change in sectoral labor allocation. Progressivity is the only instrument considered that combines redistribution with a substantial increase in formality. At equal revenue it yields three to four times the welfare gain of either level instrument, and its welfare gains persist when transfers are held fixed. Greater earnings inequality increases the welfare gains from progressivity while changing its optimal level relatively little.

These findings concern the household labor-supply margin. Firms' formalization decisions and income pooling across household members provide additional channels through which taxes can affect informality. The earnings-risk calibrations show how the balance between persistent and transitory risk shapes self-insurance and the gains from redistribution. Both alternative calibrations imply optimal progressivity several times above their benchmarks, with smaller welfare gains than in the baseline. The results show that a large informal sector leaves substantial scope for redistribution when tax design raises the return to formal work at the bottom.

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A Data Sources

In this appendix, we discuss the sources of information to construct labor market indicators and inequality, and the details entailed in each general definition. Overall, our main sources of information are household surveys for Brazil, Colombia, Mexico and Peru. In what follows, we describe some considerations for the construction of the statistics used throughout the paper. In all countries, we restrict our samples to people age 15 and older.

A.1 Brazil

The main source of data for the analysis of labor market outcomes over the income distribution is the *Pesquisa Nacional por Amostra de Domicilios Continua* (PNADC) carried out by the Brazilian Statistics Bureau (IBGE). The PNADC is a nationally representative survey of Brazilian households that provides information of the characteristics of the households and their members, and their labor market status. All households in the PNADC are interviewed by five consecutive quarters. The current version of the PNADC has run continuously since 2012, and the data are publicly available on the IBGE website.

Throughout the paper (see Section 2), we use information of labor force participation, employment, and unemployment, formality status (formal or informal) and labor earnings. Labor force status follows standard classification used by the IBGE and is available for all people age 14 and older.[15] For those employed, the survey asks questions to characterize the job. We proxy formality status based on a definition used by the IBGE (IBGE, 2020). Using the worker's main job, a worker is considered a formal-sector worker if he or she works for a person or a firm and reports having a registered job, which in Brazil is mandatory and implies having a work card (*carteira de trabalho*) signed by the workers' employer. For self-employed workers and employers, we define the formality status based on whether they have their businesses registered with the tax authorities (CNPJ).

The PNADC also collects information about the workers' labor earnings. Earnings are self-reported, cover all jobs (including salaried and independent work), and refer to monthly earnings. Earnings are in current Brazilian Reais (BRL$).

[15] The definitions among countries are very similar and resemble those used by the US Current Population Survey.

A.2 Colombia

The source of data for Colombian labor market is the *Gran Encuesta Integrada de Hogares* (GEIH) carried out by the Colombian Statistics Bureau (DANE). The GEIH is a national representative survey that collects information of labor market outcomes and social conditions of the households. It is the official source of employment statistics, incidence of monetary poverty and income inequality. The GEIH is a repeated cross section dataset collected monthly and has run continuously since 2008. Data are publicly available in the DANE website.

We define the labor force status following DANE's standard definitions, which are available for all people age 12 and older (10 and older in rural areas). We use DANE's official definition for informal employment, which defines an informal worker as a person who either works in small production units (up to five workers), excluding public-sector workers and professional self-employed (DANE, 2009). This definition applies for all workers. Although the official statistics is reported for urban areas only, we also apply the definition of informality for workers in rural areas.

Regarding labor earnings, we use monthly labor earnings from the main job, reported in Colombian Pesos (COP$).

In contrast to the other countries, the GEIH is a repeated cross section dataset, which implies that is not possible to follow individuals over time to get labor market transitions by comparing individuals over time. Nonetheless, the GEIH includes retrospective questions (e.g., tenure on the job, the type of job the person had before, how many weeks an unemployed person has been looking for a job and so on) that allow to infer labor transitions.

A.3 Mexico

The main source of data for Mexico is the *Encuesta Nacional de Ocupación y Empleo* (ENOE) carried out by the Mexican Statistics Bureau (INEGI). The ENOE is a nationally representative survey of Mexican households used to characterize their labor market outcomes. The survey is the official source of labor force statistics. In contrast to other datasets used in this analysis, the ENOE is not the dataset used to measure monetary poverty and income inequality. As in the case of Brazil's PNADC, the ENOE interviews all households by five consecutive quarters. The ENOE has run continuously since 2005. Data are publicly available on the INEGI website.

In the information of labor market status (employed, unemployed, out of the labor force) we follow the classification elaborated by INEGI. The data about labor market outcomes are available for all people age 12 and older, however, official statistics are reported for people age 15 and more

only. Regarding the informality definition, the ENOE contains a variable to classify workers between formal and informal employment. As in the case of Colombia, the definition in Mexico follows the guidelines proposed by the International Labor Organization for the measurement of informal employment (ILO, 2003). Broadly speaking, an informal worker is a person working in a job outside the scope of government regulation – and therefore is unprotected by labor legislation – either because he or she works in a small, low-productivity unit (typically unincorporated, including self-employment and subsistence activities), his or her job are uncovered by the law, or their employer keep his or her job off-the-books (INEGI, 2014). Compared to Colombia and Brazil, the operational definition of informality in Mexico is more detailed, as it includes specific questions to characterize the features of an informal job.

We use monthly labor earnings from the main job, reported in Mexican Pesos (MXN$).

In addition to the cross sections, the results in Section 2.3 and the calibrations of Appendix C exploit the ENOE's rotating-panel design, linking households and their members across the sixteen quarters from 2021Q1 to 2024Q4, a window chosen to fall entirely within the survey's current redesigned version. Individuals are linked with the standard household-person key augmented by the dwelling and sampling-unit identifiers, which removes the spurious matches the standard key alone produces; in the resulting file, duplicate matches within a quarter are below 0.2 percent and the sex of linked individuals is consistent in essentially all pairs. The linked panel contains about 5.1 million person-quarter observations on 1.6 million individuals. The persistence estimates use log hourly earnings from the main job for formal salaried workers aged 25 to 60, roughly 677,000 wage observations on 318,000 workers, and fit a persistent AR(1) component plus serially uncorrelated noise to the autocovariances at lags zero to four, following the covariance-structure approach of Abowd and Card (1989). All panel statistics use the survey's sampling weights, and the construction is documented in the replication package.

Informal returns to earning power: the switcher regression

Equation (1) relates the formal and informal hourly wages of a worker with given earning power. We estimate it on workers aged 25 to 60 whom the linked panel observes in informal work at least once and in formal work at least twice within their five interviews. Wages are log hourly earnings from the main job, trimmed at the 0.5th and 99.5th percentiles and residualized on a quadratic in age, sex, a quadratic in years of schooling, and state, locality-size, and quarter dummies.

Three problems rule out simpler estimators. Comparing the dispersion of earnings across sectors

uses different people in each sector, and is contaminated by exactly the selection on earning power that the model is about; it returns a value above one. Regressing a switcher's informal wage on their formal wage by OLS compares the same worker across sectors, but is attenuated by measurement error in survey earnings (Bound et al., 2001) and returns 0.36.[16] The usual switcher regression, with a worker fixed effect common to both sectors and a sector dummy (e.g. Bargain and Kwenda, 2014; Engbom et al., 2022), identifies a level gap but de facto imposes $\xi = 1$: earning power raises formal and informal wages one-for-one.

We instead let the return to persistent earning power differ by sector, as Lemieux (1998) and Gibbons et al. (2005) do for union status and for industries and occupations. We estimate (1) instrumenting the worker's formal log wage with a *second* formal observation of the same worker at a different date, thus $\hat{\xi} = \operatorname{cov}(\ln W^s_{it}, \ln W^f_{it''})/\operatorname{cov}(\ln W^f_{it'}, \ln W^f_{it''})$ with $t'' \neq t'$, which removes attenuation from serially uncorrelated measurement error under the assumption that selection into switching does not depend on innovations to earning power around the switch, so that among switchers the wage covariances keep the form implied by the estimated persistence.[17] Because earning power mean-reverts between interviews, the two covariances are comparable only when the informal and the formal observation are equally distant from the instrument; when they are not, we rescale by the estimated quarterly persistence of residual wages, 0.970, and we also report the lag-matched subsample in which no rescaling is needed.[18] Standard errors are from a bootstrap clustered at the worker level, and all estimates use the survey's sampling weights.

The estimand is the elasticity for workers observed changing sectors within the panel. Selection on the level of earning power changes the distribution of the regressor, while selection on innovations to earning power at the time of a sector change can affect the validity of the instrument, the concern Gibbons et al. (2005) raise when workers learn about their earning power and change

[16] In an earlier Mexican urban panel, Gong and van Soest (2002) likewise attribute much of the quarter-to-quarter variation in reported wages to measurement error.

[17] Values of a variable at other dates are valid instruments when its measurement error is serially uncorrelated (Griliches and Hausman, 1986) and, here, when the two formal observations share no persistent component other than earning power. A premium tied to formal employment, such as an employer wage premium lost on moving, would violate this and bias the estimate downward. We find no sign of such a premium: the estimate is 0.82 (s.e. 0.04) when the two formal observations lie on opposite sides of an informal spell, and so in different formal spells, and 0.85 (s.e. 0.03) when they lie on the same side.

[18] A persistent component of earning power that both sectors pay equally biases the estimate toward one. Let $\ln W^f_{it} = Z_i + \ln h_{it} + \nu^f_{it}$ and $\ln W^s_{it} = a + Z_i + \xi \ln h_{it} + \nu^s_{it}$, with Z_i uncorrelated with h_{it} and with sector choice, like the component valued equally across sectors in Lemieux (1998) and Gibbons et al. (2005). The error term of (1) then also contains $(1-\xi)Z_i$, which is correlated with every formal observation of the worker. For lag-matched triples in which the instrument is k quarters from the other two observations, the estimate converges to $(\operatorname{var} Z_i + \xi c_k)/(\operatorname{var} Z_i + c_k)$, where c_k is the covariance of $\ln h_{it}$ between dates k quarters apart. This lies between ξ and one, so for $\xi < 1$ the lag-matched estimate overstates ξ, and the rescaled estimate approximately so.

sector in response. The interpretation of the estimate as the structural curvature therefore requires the stated condition on selection into switching. The estimate depends somewhat on when the instrument is observed: it is 0.88 (s.e. 0.04) when the instrument predates both the informal and the other formal observation, and 0.79 (s.e. 0.03) when it follows both. The difference does not come from the mean-reversion adjustment, which leaves both estimates unchanged when it is replaced by the observed autocovariances of formal wages, and every interval excludes one.

Table A.1: Estimates of the curvature of informal returns, ξ

	Hourly wage		Monthly earnings	
Informality definition:	Baseline	Soc. security	Baseline	Soc. security
Cross-sectional variance ratio	1.048	1.048	1.232	1.232
Same-worker OLS	0.355	0.355	0.413	0.411
Same-worker IV	0.840	0.832	0.852	0.843
	(0.027)	(0.027)	(0.027)	(0.026)
Same-worker IV, lag-matched	0.827	0.818	0.866	0.859
	(0.036)	(0.030)	(0.036)	(0.031)
Joint IV, with informal self-employment	0.784	0.747	0.789	0.785
	(0.022)	(0.024)	(0.021)	(0.023)
Joint IV, salaried, lag-matched	0.813	0.805	0.865	0.858
	(0.037)	(0.036)	(0.034)	(0.032)

Notes: ENOE linked panel, 2021Q1–2024Q4, workers aged 25–60. Regressions relate informal to formal log earnings. The "Same-worker" rows use earnings residualized on observables: 18,990 switchers contribute 102,102 wage triples, including 14,960 lag-matched triples. IV uses a second formal observation of the same worker; lag matching places both wage observations equally far from the instrument. "Baseline" uses the ENOE formality classification; "Soc. security" classifies jobs without social-security coverage as informal, so almost all self-employment is informal under it (99.8% of own-account workers and 99.4% of employers, against 87% and 55% under the baseline). The penultimate row adds informal self-employment (31,048 switchers, 167,092 triples; 23,766 and 130,444 under the social-security definition) and jointly estimates the slope and intercept; the last row applies the same joint estimator to salaried workers and lag-matched triples (11,992 switchers, 15,011 triples; 11,894 and 14,886). Sample sizes are the same for hourly wages and monthly earnings. Only the "Same-worker IV" row adjusts for mean reversion. All estimates use survey weights; parentheses report worker-clustered bootstrap standard errors.

Table A.1 reports the estimates. Across wage concepts, informality definitions, and the treatment of mean reversion, the instrumented elasticity lies between 0.75 and 0.87, and every confidence interval excludes one. The lag-matched estimates agree with the rescaled ones, so the results are

robust to the mean-reversion-adjustment assumption; lag matching requires only that the covariance of earning power between two dates depend on the distance between them, which also holds if earning power is time invariant. Residualizing wages separately by sector, so that the returns to observables also differ across sectors as in Lemieux (1998) and Gibbons et al. (2005), lowers the estimate to 0.73 (s.e. 0.03). The last two rows also identify the intercept. The last row, which needs no mean-reversion adjustment, is the specification that calibration (b) of Appendix C imposes. The raw intercept depends on the units in which wages are measured, so we report it in unit-free form, as the predicted informal-to-formal wage ratio at the mean formal wage: in the first column, 0.940 (s.e. 0.005) in the penultimate row and 0.921 (s.e. 0.007) in the last row, evaluated at the mean over all formal workers, and 0.949 (s.e. 0.005) and 0.930 (s.e. 0.007) at the mean over switchers. The social-security definition gives 0.949 and 0.922 at the mean over all formal workers; monthly earnings give lower ratios, 0.865 and 0.871 (0.868 and 0.869 under the social-security definition), because informal main jobs have fewer hours. The scale parameter Ξ is not separately identified, since the intercept also contains the equilibrium prices w^s and w^f, which the survey does not observe; the model computes the same wage ratio at its own prices, and calibration (b) of Appendix C targets the last row's ratio at the mean over all formal workers, 0.921. Because survey earnings are net of taxes, that calibration scales the last row's slope by $1 - \tau^P$ to obtain the curvature with respect to gross earnings, $\xi = 0.80$.

An independent pattern in the same data corroborates $\xi < 1$ without relying on the instrument. Relative to formal workers who start from the same wage decile and stay formal, the wage loss one quarter later of those who move into informality is between 0.07 and 0.10 log points from the second to the sixth decile and then grows steadily to 0.35 log points in the top decile. If informal earnings were proportional to earning power, the loss would not depend on where in the distribution the worker starts; with $\xi < 1$ it should widen by $(1 - \xi)$ times the gap in earning power between deciles. Formal wages differ by 1.8 log points between the second and the top decile, about half of which is persistent, so ξ between 0.78 and 0.84 predicts a widening of 0.14–0.20 log points; the observed widening is 0.26. Fixed-effects quantile estimates for Mexico, Brazil, and South Africa find instead that the informal penalty for salaried workers is larger in the lower part of the conditional wage distribution and tends to disappear at the top (Bargain and Kwenda, 2014), a pattern Haanwinckel and Soares (2021) treat as a stylized fact. Their worker fixed effect is common to both sectors and so imposes $\xi = 1$; the quantiles then rank a worker's quarters by how far the wage sits from the worker's own average, not workers by earning power, and more volatile informal pay by itself makes

the penalty shrink toward the top of that ranking.

A.4 Peru

The main source of data for Peru is the Encuesta Nacional de Hogares (ENAHO) carried out by the Peruvian Statistics Bureau (INEI). The ENAHO is a nationally representative survey of Peruvian households used to characterize labor market outcomes, and households' living standards. It is the official source of employment statistics, incidence of monetary poverty and household income inequality. The ENAHO has a mixed structure in which a subsample of households (about 25% of total) are interviewed every year for up to five years. ENAHO data are available continuously since 2008. Data are publicly available in the INEI website.

Information about labor market outcomes is available for all people age 14 and older, and we follow the classification made by the INEI. As in the case of the ENOE in Mexico, the ENAHO includes detailed information that allow to classify an informal worker depending on the type of production unit they work and whether they are covered by labor regulation.

The ENAHO collects information about the workers' labor earnings in all jobs to compute measures of annual labor earnings. Throughout the paper, we take this measure of annual labor earnings divided by 12 to get a measure of monthly labor earnings. Earnings are in current Peruvian Soles (PEN$).

B Inequality and Informality

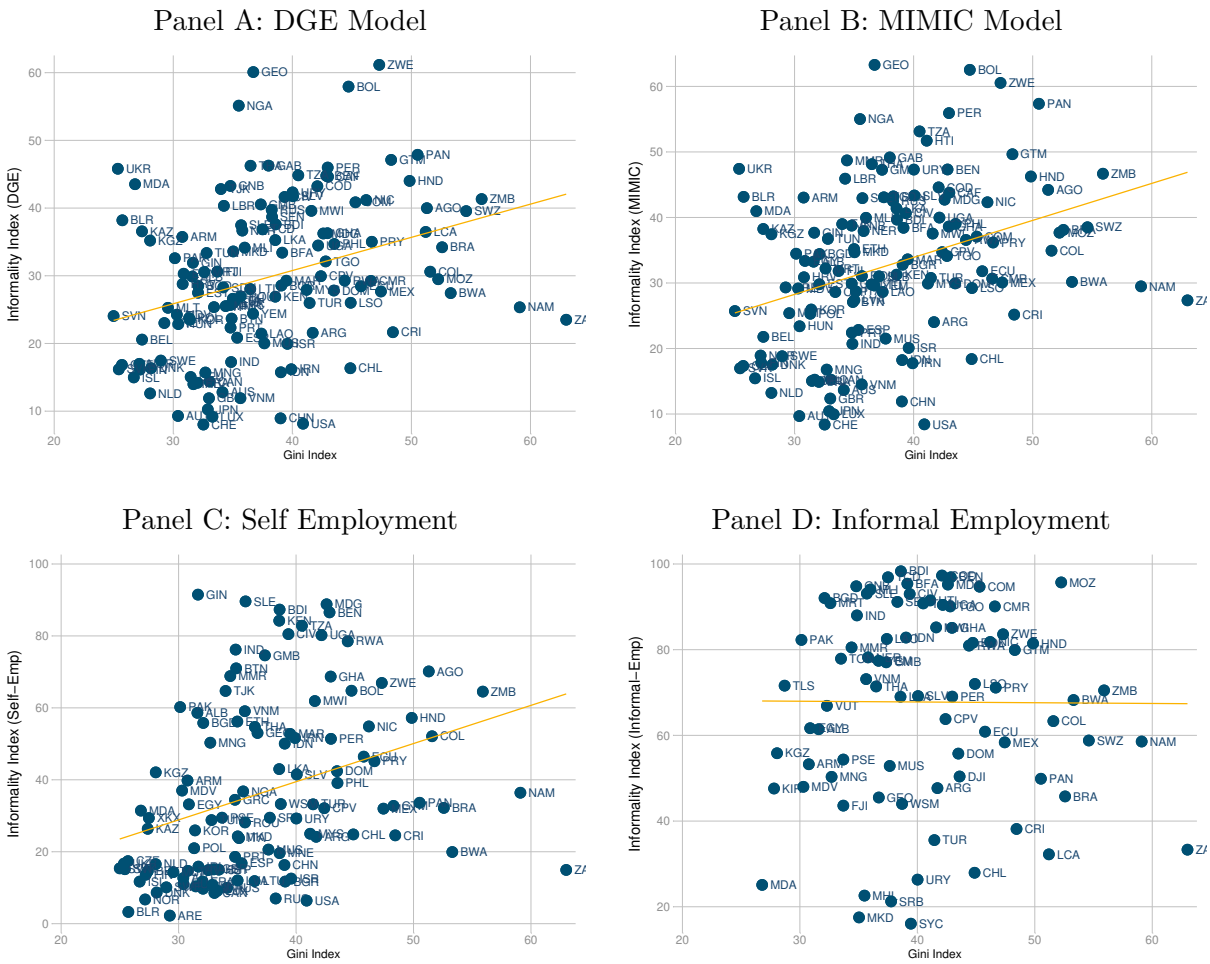
Several confounding variables, such as income levels or human capital in a country, might influence the relationship between inequality and informality. A thorough analysis of this relationship is beyond the scope of this paper; here, we simply compare the relationship between the Gini index of income inequality reported by the World Bank with informality measures produced and updated by Kose et al. (2021). Given its multidimensional nature, we employ multiple direct and indirect estimation methods to account for the difficulty in measuring informality. We focus on measures with sufficient variation across countries and time.

The analysis includes two indirect measures: The general equilibrium model (*DGE*) and the multiple indicator multiple causes model (*MIMIC*). The *DGE* model estimates the size of the informal economy as a percentage of GDP using the approach in Elgin and Oztunali (2012), which is available for 158 countries. The *MIMIC* model measures the size of the informal sector as a percentage of GDP using the model of Schneider et al. (2010), available for 160 countries.

We also include two direct measures of informality based on country surveys. The first measure is the share of self-employment in total employment (*Self-Emp*). This category encompasses four sub-categories of jobs: employers, own-account workers, members of producers cooperatives, and contributing family workers. The share of *Self-Emp* is obtained from the labor force and household surveys for a sample of 180 countries. The second direct measure corresponds to the share of informal employment in percent of total employment (*Informal-Emp*), obtained from household and labor surveys and available for 72 countries.

Figure B.1 presents the cross-sectional scatter plot between the Gini index average and the four informality measures from 2002 to 2022. We observe a positive correlation between inequality and the indirect measures of informality and the share of self-employment in Panels A to C. There is little correlation between inequality and the measure of informal employment in Panel D. The number of observations for this measure is also smaller.

Table B.1 reports panel regressions with and without country fixed effects. Both indirect measures remain positively associated with inequality. With country effects, the self-employment coefficient becomes small and statistically insignificant, while the informal-employment coefficient becomes statistically significant.



Notes: The figure compares countries using data from 2002–2022. The Gini index is on a 0–100 scale. DGE and MIMIC measure informal output as a percentage of GDP; Self-Emp and Informal-Emp measure self-employment and informal employment as percentages of total employment. Sources: World Development Indicators (World Bank) and Kose et al. (2021).

Figure B.1: Cross-sectional correlations between inequality and informality

Table B.1: Panel Regressions of Informality on Inequality, 1990–2022

	Informality Measures							
	Indirect				Direct			
	DGE		*MIMIC*		*Self-Emp*		*Informal-Emp*	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Gini Index	0.636***	0.222***	0.688***	0.215***	1.017***	0.105	0.415	0.939**
	(0.13)	(0.05)	(0.14)	(0.03)	(0.14)	(0.12)	(0.33)	(0.42)
Observations	1746	1746	1673	1673	1530	1530	329	329
Country FE	No	Yes	No	Yes	No	Yes	No	Yes

Notes: Each column reports the coefficient on the Gini index in a regression of the indicated informality measure, with country fixed effects where shown. DGE and MIMIC measure informal output relative to GDP; Self-Emp and Informal-Emp measure self-employment and informal employment relative to total employment.
Source: World Development Indicators (World Bank) and Kose et al. (2021).

C Robustness Analysis

This appendix asks whether the welfare-maximizing progressivity of Section 5, about six-and-a-half times the calibrated benchmark, depends on two choices of the baseline calibration that the targeted moments do not settle: the income process, which the baseline parameterizes rather than estimates (Section 4), and the curvature of informal returns to ability ξ, which the calibration fits at 0.92 while our panel estimates lie between 0.75 and 0.87 (Appendix A.3). We re-calibrate the model replacing each choice with its ENOE-panel counterpart, one block at a time: alternative calibration (a) adds the sector-specific transitory earnings risk measured in the panel, alternative calibration (b) fixes ξ at the panel estimate and imposes the panel's earnings persistence. Each economy is re-fitted to the baseline's targeted moments, and each subsection compares the fit and policy results with the baseline. In both, the ratio of the welfare-maximizing progressivity to the benchmark is of the same order as in the baseline: about three in calibration (a) and about eight in calibration (b).

C.1 Alternative calibration (a): sector-specific transitory earnings risk

Using the ENOE panel, we estimate each sector's cross-sectional variance of log earnings and decompose it into persistent and transitory components: of the formal variance of 0.325, 0.132 is transitory; of the informal 0.674, 0.313 is. The transitory component is 2.4 times larger in the informal sector, and roughly half the sectoral variance gap is transitory rather than persistent: of the gap of 0.349, 0.180 is transitory, and the baseline targets only its persistent component, 0.169. Alternative calibration (a) adds the measured sectoral variances to the baseline through an i.i.d. multiplicative earnings shock driven by a single common factor with sector-specific loadings, so that $\mathrm{Var}(\ln \varepsilon)$ matches the panel value in each sector and the transitory component of informal earnings is more volatile, with the ergodic dispersion of the persistent component σ_{LR} left free for the moment set to determine.[19] Every other ingredient is the baseline's, and the targeted moments are those of Section 4.7, except that the variance-gap target is the total gap, 0.349, rather than its persistent component, since the model's sectoral earnings now include the transitory component.

With the transitory layer imposed, the calibration chooses $\sigma_{LR} = 0.785$, below the baseline's 1.00, matches the total sectoral variance gap (0.349 against 0.349), and places the curvature of informal returns at an interior $\xi = 0.98$. It also lowers the productivity of informal relative to

[19] The panel identifies the two sectoral transitory variances but not their correlation, which must be chosen. We use a single common factor with sector-specific loadings (correlation one). With uncorrelated draws across sectors, hedging the draw becomes the first-order motive of the allocation of labor across sectors, dwarfing the inclusion and evasion margins through which progressivity operates, and the welfare profile over progressivity is flat.

formal work ($\varkappa/\Omega = 0.62$, against 0.88 in the baseline) and raises the vacancy cost about sevenfold, so that the labor agency's surplus on a formal match is much larger relative to the informal wage than in the baseline (0.44 against 0.05). Three other aspects of the fit deteriorate. First, wealth and consumption inequality and the capital-output ratio fall below their data targets (Gini_w 0.66 against 0.72, Gini_c 0.37 against 0.41, K/Y 2.31 against 2.44), because transitory shocks are self-insured through the asset buffer and are less effective than persistent ones in generating precautionary savings. Second, informality is below the data, with an informal headcount of 0.49 against 0.54 (an informal share of hours of 0.50 against 0.51), while the job-finding rate is above it (0.18 against 0.16) and the mean log-earnings gap falls to 0.64 against 0.72, and the informal-to-formal net wage ratio falls to 0.72 against 0.92. The added transitory component improves the fit of earnings dispersion but makes the two labor-market targets harder to match jointly. Third, no worker splits hours between the two sectors, against 4.0% in the baseline and 2.4% of workers holding jobs in both sectors in the data. The common factor is approximated with a five-node Gauss–Hermite quadrature.

Optimal progressivity remains well above the benchmark after adding the estimated transitory component, although by less than in the baseline. This calibration's benchmark progressivity is 0.018, and the welfare-maximizing level is $\tau^P = 0.060$, about three times the benchmark, under *all three* budget closures. The gain at the optimum is 1.85% of consumption under the benchmark closure (1.26% under fixed absolute spending and 0.58% with the transfer fixed, where the optimum is interior and leaves G/Y above its benchmark level), and 1.12% when leisure is excluded. As in the baseline, the optimum coincides with the peak of tax revenue as a share of GDP. Unlike in the baseline, informality does not fall: the informal share of hours rises with progressivity throughout the explored range and is 0.3 percentage points above its benchmark level at the optimum. The reason lies in the benchmark allocation described above: almost every worker with a formal job already works only formal hours, so progressivity has no informal hours to draw into formal work, the margin behind the fall in informality in the baseline. The small increase reflects formal workers reducing their hours by about 1% and a slightly lower job-finding rate, which responds about a third as much as in the baseline because the surplus on a formal match is large. The gain therefore comes from insurance and redistribution, with consumption inequality falling by 1.6%, rather than from formalization. The lower optimum is in line with the higher ξ this calibration selects: as Appendix D.5 shows, informal returns closer to linear in ability make the evasion margin bind at lower progressivity.

C.2 Alternative calibration (b): informal returns fixed at the panel estimate

Alternative calibration (b) uses the panel estimates to fix informal-return curvature and earnings persistence, and targets the panel's informal-to-formal wage ratio (Appendix A.3). The joint slope–intercept specification on lag-matched observations of salaried workers gives an instrumental-variables elasticity of 0.813 (s.e. 0.037). Since survey earnings are net of the labor tax, the corresponding curvature with respect to gross formal earnings is $0.813\,(1-\tau^P) = 0.80$, evaluated at benchmark progressivity. We fix the model's ξ at this value throughout the policy experiments. The persistence of the efficiency process is set to the panel's annual estimate, $\rho = 0.944$, instead of the baseline's 0.98, and σ_{LR} is re-calibrated. This calibration combines the panel estimates of ξ and ρ; Appendix D.5 reports the effect of moving ξ alone, without re-calibration.

With ξ fixed below one and σ_{LR} free, the calibration keeps much of the baseline inequality fit: σ_{LR} comes back at 1.02, wealth inequality is close to its target (Gini_w 0.70 against 0.72), consumption inequality is somewhat lower (0.38 against 0.41), K/Y is close (2.38 against 2.44), and the model's formal log-earnings variance (0.328) is close to the panel's total (0.325, of which 0.193 is persistent). The fit of sectoral earnings and of informality deteriorates. With ξ well below one and a single persistent process common to both sectors, informal earnings compress ability differences rather than amplifying them, so the model can generate an informal variance excess only through sorting and the intensive margin: the persistent variance gap falls to 0.11 (baseline and data 0.17), the mean log-earnings gap overshoots at 0.98 (baseline 0.71, data 0.72), and the targeted wage ratio itself cannot be reached (0.85 against 0.92). The wage ratio trades off against informality: low-productivity workers leave formal work, so the formal pool selects on high productivity, where the informal-to-formal ratio is lower when $\xi < 1$, and the formal wage premium set by the wage rule widens. The calibration reaches a ratio of 0.85 only at an informal share of hours of 0.75, against 0.51 in the data and 0.58 in the baseline; with σ_{LR} near one, a ratio of 0.92 requires an informal share of hours of 0.9 or more. The panel explains why the variance gap is hard to reach. In the panel, the excess persistent dispersion of informal earnings is concentrated among the self-employed, whose earnings pool labor and business income. The informal variance excess in the data is thus a composition effect between two groups, not a property of how a single informal technology maps ability into earnings, and a model with one informal technology cannot carry it once ξ is fixed at the panel estimate.

This calibration also favors substantially higher progressivity. Its benchmark progressivity is 0.016, and the welfare-maximizing level under the benchmark closure is $\tau^P = 0.125$, about eight

times the benchmark, with a CEV of 1.24% (0.090 and 0.43% under the fixed-absolute-spending closure, and 0.110 and 1.19% with the transfer fixed and G/Y at least at its benchmark level). That the optimum lies above the baseline's 0.100 is consistent with the sensitivity analysis of Appendix D.5, where a lower ξ raises the optimum. Informality falls at the optimum, by 7.2 percentage points of the informal share of hours, consumption inequality falls by 2.8%, and measured TFP rises by 0.28%, so the qualitative content of Section 5.1 is unchanged. Two features differ from the baseline. The welfare profile is flat at the top: the CEV varies by less than 0.03 percentage points between $\tau^P = 0.110$ and 0.135, so the location of the optimum within that range is not sharply determined. And tax revenue as a share of GDP peaks at a lower progressivity, $\tau^P \approx 0.06$, and is 0.15 percentage points below its benchmark level at the optimum. The gains are also smaller than in the baseline (4.34%).

D Additional Model Results

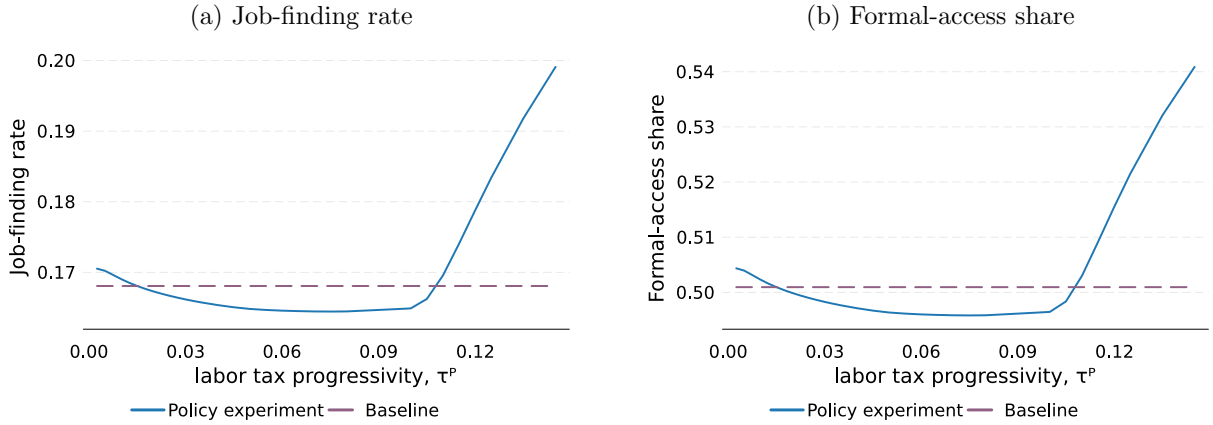
This appendix reports additional results from the quantitative model. Section D.1 reports how job finding and access to formal jobs respond to progressivity. Section D.2 examines the robustness of the optimal-progressivity results to the government budget closure, repeating the main experiment under two alternatives to the benchmark (constant G/Y, transfer-rebated) closure. Section D.4 states the welfare criterion and the consumption-equivalent metric used throughout. Section D.5 reports the sensitivity of the optimum to the wage-rule, labor-supply, and informal-returns parameters. Section D.6 presents results from an economy with homogeneous labor-market transitions, showing that the negative informality profile over the income distribution persists even without skill-specific transition rates. Section D.7 reports the informal share of hours worked by labor-income decile and compares the model's average tax rates, including the tax on business profits, with official incidence estimates. Section D.8 reports comparative statics from the analytical model.

D.1 Labor-Market Frictions and Access to Formal Jobs

Progressivity also changes access to formal jobs, through the vacancy posting of the labor agency. Figure D.1 reports the average job-finding rate into the formal sector and the share of households holding a formal job along the progressivity experiment of Section 5.1. Between the baseline and the optimum, $\tau^P = 0.10$, both are nearly flat: the job-finding rate moves from 0.168 to 0.165, and the formal-access share from 0.501 to 0.496. Up to the optimum, the most productive households keep all their hours in formal work, so formal labor and the labor agency's surplus change little, and the fall in informality comes from the inclusion margin among households that already hold formal jobs.

Beyond the optimum, the evasion margin changes the agency's incentives. As high-productivity households move hours into informal work, effective formal labor becomes scarcer, its price rises, and the agency's surplus per efficiency unit of formal labor, $\Omega p^f - w^f$, increases by 63% between the optimum and the top of the grid, $\tau^P = 0.145$. Vacancy posting responds: the job-finding rate rises by about 21%, and the formal-access share by 4.4 percentage points. Job finding rises for every worker without formal access, so the new formal jobs are spread roughly in proportion to them: households in the bottom 75% of the productivity distribution, 79% of those without access, take 77% of the new jobs. Each newly attached worker supplies most of its hours formally, nearly all of them in the bottom group and about 80% among top earners, so one newly attached top earner

offsets the reallocation of about four top earners already in formal jobs. Over this range, wider access lowers the informal share of hours by 4.3 percentage points, more than offsetting the 2.9 points added by the top earners' reallocation, so the informal share keeps falling, by 1.1 points in total, even as evasion erodes the formal tax base and revenue falls.



Notes: Labor-tax progressivity τ^P varies as in Figure 7: government consumption remains 15% of output, the lump-sum transfer balances the budget, and other tax parameters are fixed. Panel (a) reports the average rate at which households without a formal job find one, $f(\theta)$. Panel (b) reports the share of households holding a formal job, entrepreneurs included. Dashed lines mark the baseline.

Figure D.1: Job finding and access to formal jobs as progressivity varies

D.2 Alternative Government Budget Closures

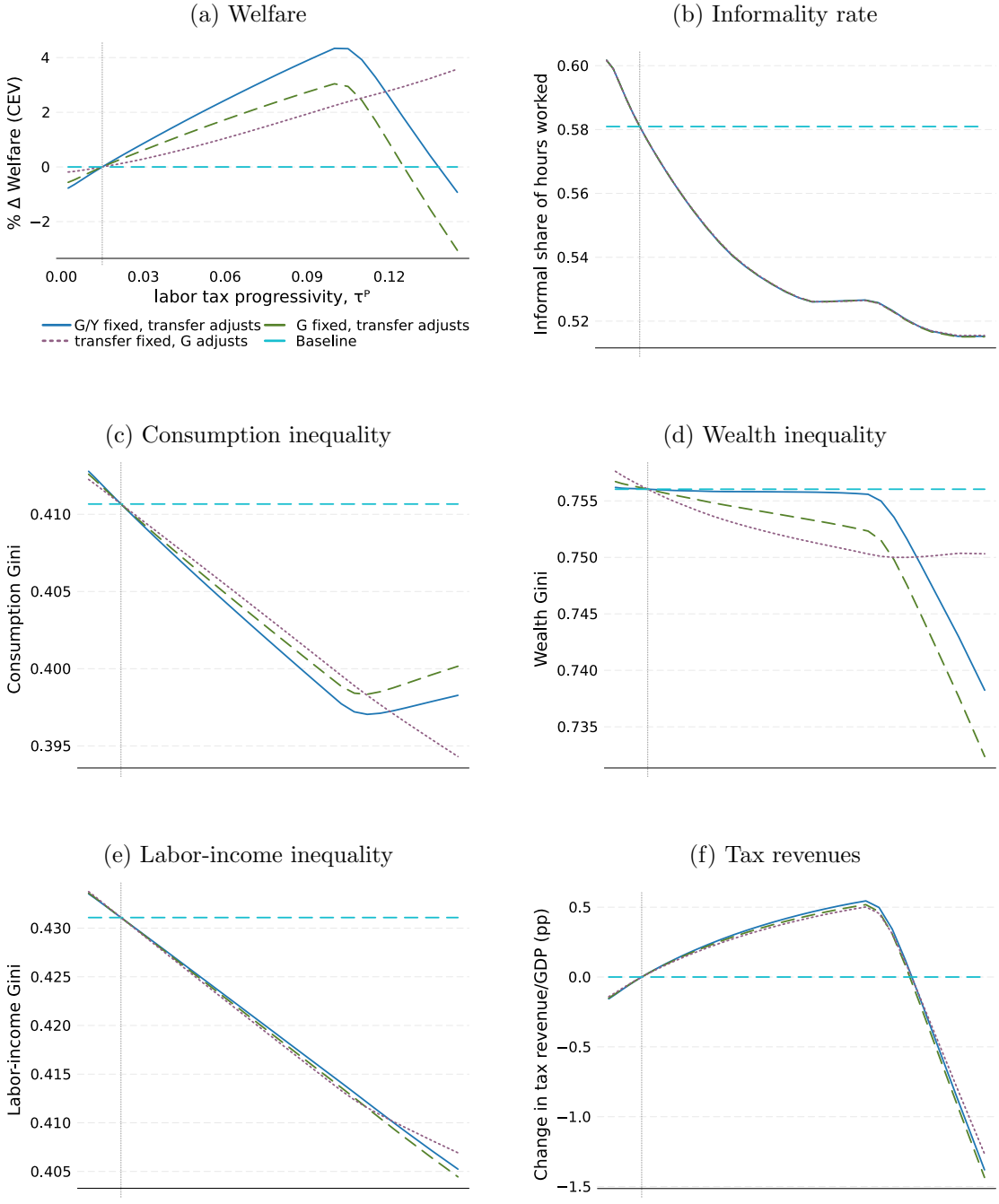
The benchmark experiment holds the spending-to-output ratio at its calibrated $G/Y = 15\%$ and returns changes in revenue net of spending to households through the lump-sum transfer. This appendix repeats the progressivity experiment under two alternative closures. In the first, government consumption is held fixed in absolute terms while the transfer continues to adjust; spending then no longer scales with output, so reforms that expand (contract) the economy are not partially absorbed (cushioned) by spending. In the second, the transfer is held fixed at its baseline level and government consumption absorbs revenue changes one for one, so that households receive none of the fiscal dividend of a reform. Figure D.2 reports the resulting profiles alongside the benchmark ones, and Section D.3 extends the comparison to the other fiscal instruments.

Optimal progressivity is similar across all three closures: the welfare-maximizing progressivity is $\tau^P = 0.100$ under fixed absolute spending and $\tau^P = 0.115$ under the fixed-transfer closure, against 0.100 in the benchmark, and informality at the optimum falls to within a fraction of a percentage

point of its benchmark-closure value in every case. The informality path itself is practically the same under the three rules (Panel D.2b): sector choice responds to the marginal returns set by the tax schedule, while the rules differ only in how revenue is used, as a lump-sum transfer or as government consumption, and the income effects of that difference on labor supply and sector allocation are small. At the top of the grid, the transfer and government consumption differ across the rules by up to 1.9 and 1.7 percentage points of output, yet aggregate labor supply differs by less than 1% and the informal share of hours by 0.04 percentage points. Under the fixed-transfer closure, government consumption absorbs every change in revenue and has no utility value, so welfare keeps rising along the explored grid as government consumption shrinks. We therefore report this closure's optimum subject to G/Y not falling below its benchmark value of 15%. This spending requirement limits the reform: G/Y falls below 15% at $\tau^P \approx 0.12$. Requiring in addition that government consumption not fall in levels gives $\tau^P = 0.105$ and a gain of 2.39%, while requiring in addition that output not fall below its benchmark level leaves the benchmark schedule as the best one: output falls at every progressivity above the benchmark, and G/Y falls below 15% at every progressivity below it. What the closure changes is the size of the gains: 4.3% of consumption in the benchmark, 3.0% with absolute spending fixed, and 2.7% when government consumption adjusts and transfers are fixed. The ordering is instructive. Under the fixed-transfer closure households receive none of the fiscal dividend of formalization, so gains remain even without additional lump-sum resources; the closure comparison shows how the welfare gain depends on the fiscal adjustment accompanying the change in the schedule. Because the location of the optimum is closure-robust while its value is not, we regard the benchmark closure, the budget rule the calibration itself is computed under, as the appropriate basis for the optimal-policy statements in the main text. The transitory-risk calibration of Appendix C passes the same test: its optimum sits at $\tau^P = 0.060$ under all three closures, with gains of 1.85%, 1.26%, and 0.58%, and with the transfer fixed its optimum is interior and leaves G/Y above its benchmark level.

D.3 Fiscal Instruments Under Alternative Closures

Section 5.2 compares progressivity with the two level instruments, the labor-tax level and the capital tax, under the benchmark closure. Table D.1 reports each instrument's welfare-maximizing reform under all three closures, evaluated at that optimum: the size of the reform, the welfare gain, and the responses of informality, consumption inequality, tax revenue, and the budget item that clears the government budget.



Notes: Progressivity varies under three budget rules: government consumption fixed at 15% of output with the lump-sum transfer adjusting; government consumption fixed in levels with the transfer adjusting; and the transfer fixed with government consumption adjusting. Other tax parameters are fixed. The rules coincide at baseline progressivity $\tau^P = 0.015$ (vertical line); horizontal lines mark baseline outcomes. Welfare gains are steady-state CEVs (percent), including leisure. Panels (b)–(e) show levels. Panel (f) shows changes in tax revenue/GDP in percentage points. Government consumption has no direct utility value. Under the fixed-transfer rule, government consumption falls below 15% of output for τ^P above about 0.12, and welfare keeps rising as it shrinks; the text reports this rule's optimum subject to $G/Y \geq 15\%$, $\tau^P = 0.115$.

Figure D.2: The progressivity experiment under the three government budget closures

Two patterns organize the table. First, optimal progressivity stays close across budget rules: the optimum is +8.5 percentage points of τ^P under both rebate closures and +10.0 when the transfer is fixed and the spending requirement limits the reform; the informal share of hours falls by 5.4–6.0 percentage points in every case, and the welfare gain runs from 4.3% with the rebate and spending indexed to output, to 3.0% with spending fixed, to 2.7% when government consumption absorbs revenue changes. Second, the level instruments raise welfare through the rebate. Under the two closures that return revenue to households, raising the labor-tax level or the capital tax finances a demogrant and raises welfare (by up to 6.3% and 1.6% under the benchmark closure), the former at the cost of a 13–20 percentage point rise in the informal share of hours, the latter with the labor-market allocation essentially unchanged and its gains still rising at the edge of the explored range. These gains rest on much larger revenue increases than progressivity's: at the revenue change of the progressivity optimum, the labor-tax level and the capital tax yield 1.4% and 1.2% under the benchmark closure (1.2% and 0.8% with G fixed), against 4.3% (3.0%) for progressivity. Once the transfer is fixed and revenue changes accrue to government consumption, neither level tax can improve welfare by rising, and cutting either one would lower G/Y below its benchmark value: their constrained optimum is no reform, and at the change in revenue/GDP of the progressivity optimum they lower welfare, by 0.11% and 0.18%. Progressivity is the only instrument whose constrained optimum under this closure is a tax increase. Across closures, increased progressivity combines welfare gains with a substantial reduction in informality. Without the G/Y constraint, small cuts in the level taxes improve welfare only because they shrink government consumption.

D.4 Welfare Criterion

Following the quantitative literature on optimal progressivity (Conesa and Krueger, 2006; Heathcote et al., 2017; Holter et al., 2019), we evaluate reforms with a utilitarian social welfare function over the stationary distribution. Let μ_τ denote the stationary distribution over household states under progressivity τ^P, and let c and ℓ denote the associated policy functions. For the consumption-equivalent calculation, omit the additive constants in utility and define

$$\mathcal{W}^c(\tau^P) = \int \frac{c^{1-\zeta}}{1-\zeta}\, d\mu_\tau, \qquad \mathcal{W}^\ell(\tau^P) = \int \Phi \frac{(1-\ell)^{1-\phi}}{1-\phi}\, d\mu_\tau. \tag{D.1}$$

Total welfare is $\mathcal{W} = \mathcal{W}^c + \mathcal{W}^\ell$. Omitting constants leaves welfare differences unchanged and makes the consumption component homogeneous. The CEV of reform R relative to benchmark B is the

Table D.1: Welfare-maximizing reforms by instrument and budget closure

Closure	Δ instrument at optimum (pp)	Welfare CEV (%)	Δ informal hours share (pp)	Δ Gini consumption (×100)	Δ revenue / GDP (pp)	Δ budget margin (pp GDP)	CEV at equal revenue (%)
Progressivity, τ^P							
G/Y fixed	+8.5	4.34	-5.43	-1.29	+0.54	+0.58	4.34
G fixed	+8.5	3.04	-5.44	-1.18	+0.52	+0.22	3.04
Transfer fixed	+10.0‡	2.65	-6.03	-1.30	+0.11	-0.27	2.65
Labor-tax level, $1-\tau^L$							
G/Y fixed	+10.0	6.33	+19.65	-1.39	+3.28	+3.24	1.42
G fixed	+7.0	4.78	+13.29	-0.96	+2.64	+2.32	1.17
Transfer fixed	–	–	–	–	–	–	-0.11
Capital tax, τ^k							
G/Y fixed	+9.0†	1.62	-0.05	-0.24	+0.77	+0.78	1.16
G fixed	+9.0†	1.09	-0.04	-0.19	+0.75	+0.63	0.76
Transfer fixed	–	–	–	–	–	–	-0.18

Notes: Each row maximizes welfare over the explored range for one instrument under the stated budget rule; other tax parameters are fixed. The first column names what the rule holds fixed: with G/Y or G fixed, the lump-sum transfer adjusts; with the transfer fixed, government consumption adjusts. Changes are relative to the baseline. CEV denotes steady-state consumption-equivalent variation, including leisure. The labor-tax level $1-\tau^L$ is the average tax rate at mean formal earnings. The Gini column reports 100 times the absolute change. The budget margin is the change in the transfer or government consumption, divided by baseline output and multiplied by 100. Government consumption has no direct utility value. With the transfer fixed, welfare is maximized subject to government consumption remaining at least at its baseline share of output. The last column reports the instrument's CEV at the reform, nearest to the baseline, that changes revenue/GDP by as much as the progressivity optimum under the same closure (+0.54 percentage points with G/Y fixed, +0.52 with G fixed, +0.11 with the transfer fixed), interpolated linearly on the instrument's grid; with the transfer fixed, this reform is not the two level instruments' constrained optimum, which is no reform. † marks a grid boundary, not an interior optimum; ‡ marks an optimum constrained by the requirement that government consumption remain at least at 15% of output. − marks no reform: the best reform is the baseline itself. Explored ranges: $\tau^P \in [0.0025, 0.145]$, $\tau^L \in [0.755, 0.885]$, and $\tau^k \in [0.071, 0.171]$.

uniform proportional change Δ in benchmark consumption satisfying

$$(1+\Delta)^{1-\zeta}\mathcal{W}_B^c + \mathcal{W}_B^\ell = \mathcal{W}_R, \qquad \Delta = \left(\frac{\mathcal{W}_R - \mathcal{W}_B^\ell}{\mathcal{W}_B^c}\right)^{1/(1-\zeta)} - 1. \tag{D.2}$$

With $\zeta = 2$, the exponent is -1. Quantitative-model welfare figures and tables report Δ in percent, relative to the benchmark of the same fiscal closure and, in the inequality experiment, the same σ_{LR}-economy. These comparisons account for consumption and leisure in the stationary distributions; they do not include the transition between steady states.

At the welfare-maximizing progressivity of the benchmark experiment the two components move in the same direction. The optimal reform shifts hours out of informal work, about four fifths of them into formal work and the rest into leisure, as aggregate hours fall slightly, so the leisure component contributes about 1.6 percentage points of the 4.3% gain, and its contribution is flat in the neighborhood of the optimum, which is set by the consumption component.

D.5 Sensitivity: Wage Rule, Labor Supply Elasticity, and Informal Returns

Table D.2 reports the welfare-maximizing progressivity when one structural parameter at a time is moved away from its baseline value, holding all remaining parameters fixed. The optimum is insensitive to the wage-rule weight η, which governs how the surplus between the intermediate-goods price and the informal wage is split: moving η by ± 0.15 around its baseline value of 0.60 shifts $\tau^{P,*}$ by 0.005 to 0.01, so the results do not hinge on how the surplus is divided. The Frisch elasticity matters through the strength of the labor-supply response itself: doubling it to 0.60 lowers the optimum to 0.09 with larger gains, while halving it to 0.15 raises the optimum to 0.12 and halves the gain: with less responsive labor supply, progressivity has less formalization to gain, so the stakes fall, and less evasion to fear, so the planner can push the schedule further.

The curvature of informal returns to skill, ξ, moves the location of the optimum the most: lowering it by 0.05 raises the optimum to 0.135 at an unchanged gain (informal earnings compress faster at the top, so evasion is more expensive and the planner can push progressivity further), while raising it to 0.968 (informal returns closer to linear in skill) pulls the optimum down to 0.07 with about 70% of the gain, as the evasion margin bites earlier. The Frisch elasticity moves the size of the gain more than ξ does, between 2.11% and 4.99% against 3.11% and 4.35%. The sensitivity of the optimum to ξ is why ξ is disciplined directly in the calibration through the sectoral earnings moments, and why Appendix C reports two alternative calibrations that discipline ξ differently: the

transitory-risk calibration, where it settles at 0.98, and the calibration with ξ fixed at the ENOE panel estimate.

Table D.2: Sensitivity of the welfare-maximizing progressivity $\tau^{P,*}$

		G/Y fixed		G fixed	
Parameter	Value	$\tau^{P,*}$	CEV at $\tau^{P,*}$ (%)	$\tau^{P,*}$	CEV at $\tau^{P,*}$ (%)
– (baseline)	–	0.100	4.34	0.100	3.04
Wage rule η	0.45	0.095	4.11	0.095	2.86
	0.75	0.110	4.63	0.110	3.23
Frisch elasticity	0.15	0.120	2.11	0.120	1.49
	0.60	0.090	4.99	0.090	3.06
Informal curvature ξ	0.868	0.135	4.35	0.125	2.72
	0.968	0.070	3.11	0.070	2.18

Notes: One parameter varies at a time around the baseline values: wage-rule surplus share $\eta = 0.60$, Frisch elasticity 0.30, and informal-return curvature $\xi = 0.918$. Remaining parameters are not recalibrated; the Frisch elasticity is varied through the curvature ϕ of the utility of leisure, holding its weight Φ fixed. CEV denotes steady-state consumption-equivalent variation, including leisure, relative to each perturbed economy's own equilibrium at baseline progressivity $\tau^P = 0.015$. The lump-sum transfer balances the budget, with government consumption fixed at 15% of output or at its baseline level. Optima are selected over the explored progressivity grid, $\tau^P \in [0.0025, 0.145]$.

D.6 Homogeneous Labor-Market Transitions

Figure D.3 reports results for an economy in which all workers face identical formal-sector labor market dynamics. Panels D.3c and D.3d show the flat profiles of separation and job-finding rates across the labor-income distribution, confirming the absence of skill-dependent transition probabilities. As a consequence, the composition of households with and without a formal job changes, as shown in Panel D.3b: households without a formal job become more productive on average, and those with one less productive. Low-skilled households now face more favorable transition rates and the opposite happens to high-skilled ones; this improves labor productivity in the informal sector and lowers it in the formal one. In equilibrium, free entry of vacancies raises the common job-finding rate to 0.191, above the baseline average of 0.168, so the share of households with a formal job rises from 50.1% to 53.3% and the informal share of hours falls from 58.1% to 53.7%.

Even under these homogeneous labor market conditions, the model continues to generate a downward-sloping informality rate over the labor-income distribution, although a flatter one: the informal share of hours falls by about 5 percentage points per decile on average, against 8 in the

baseline and in the data. This outcome reflects the optimal labor supply and sectoral allocation implied by households' intratemporal first-order conditions (17). Sectoral labor supply responds to household-specific incentives, particularly among low-productivity workers, whose higher wealth-effect elasticities amplify the reallocation margin between formal and informal activities; households at the lowest productivity level work informally even when they hold a formal job. In contrast, the profile of informality flattens in the upper half of the income distribution, where the absence of skill-specific transition rates limits further reallocation toward the formal sector: from the fifth to the ninth decile, the informal share of hours falls by 18 percentage points, against 23 in the baseline.

D.7 Additional Fit Checks

Panel 6c in Section 4.7 measures informality as the share of workers in informal jobs. Panel D.4a measures it as the informal share of hours worked, by labor-income decile, in the baseline economy and in the data.

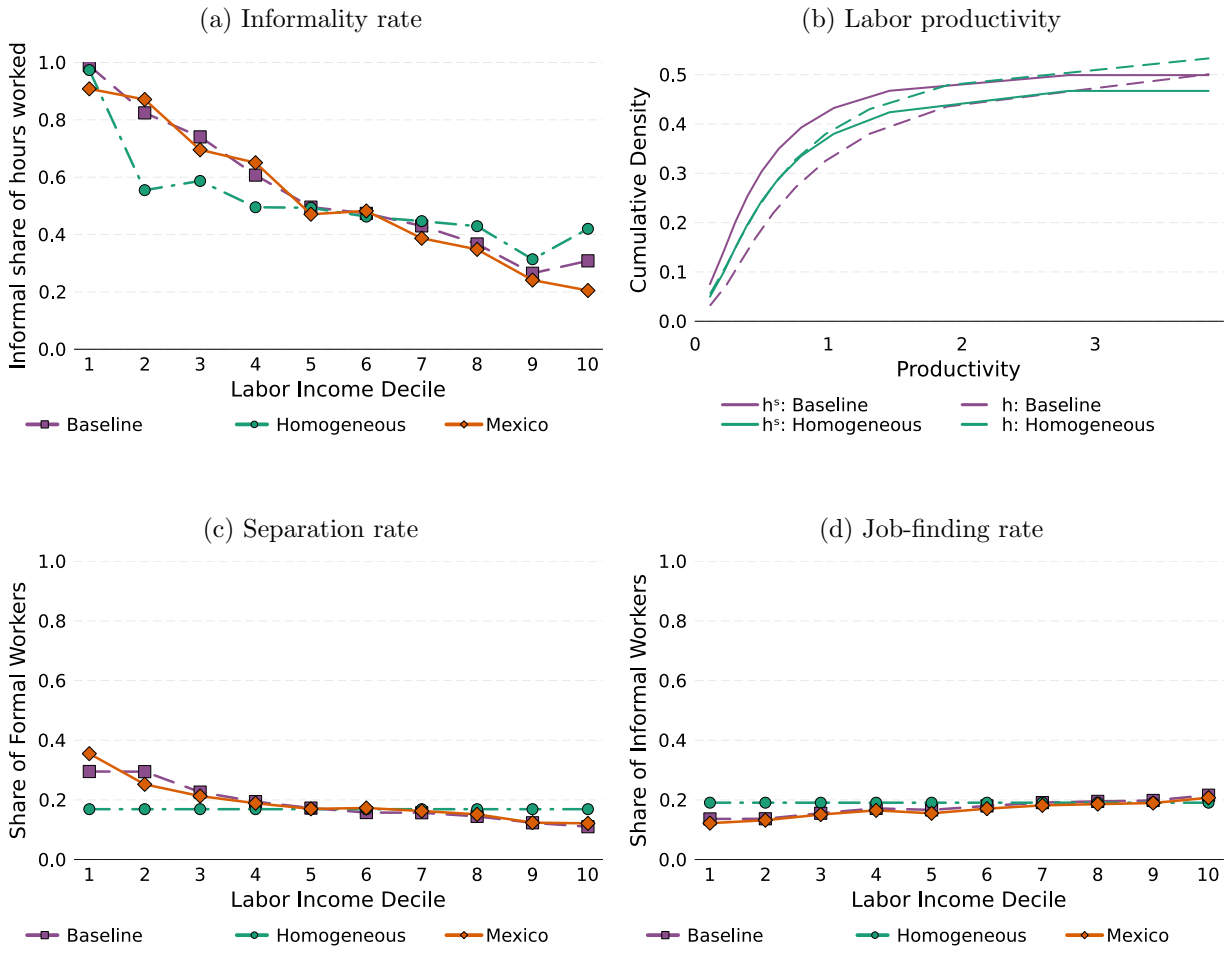
Panel 6d in Section 4.7 compares the model's average labor-tax rate by decile of household income, excluding entrepreneurs, with the incidence estimates of Mexico's Ministry of Finance (SHCP) for the income tax on salaries. Panel D.4b brings the entrepreneurs back. The model series adds the tax on the profits that entrepreneurs receive, $\tau^{\text{profit}}\,\Pi(h)$, to each household's labor tax and divides by total gross household income; the data series is the personal income tax paid by individuals, which also covers their business income, as a share of total gross household income, from the same SHCP report. Because only entrepreneurs receive profits, adding the profit tax changes the model's profile only in the deciles where the entrepreneurs are. The model tracks the data about as closely as in Panel 6d: the root-mean-squared gap across the ten deciles is 1.2 percentage points, against 1.1 there.

D.8 Comparative Statics in the Analytical Model

This section reports comparative statics from the analytical model of Section 3, examining how structural features of the economy shape optimal progressivity.

Higher public financing needs and higher unemployment risk raise informality, reduce welfare, and lower optimal progressivity. With lower spending or unemployment risk, optimal progressivity rises to 0.28–0.31.

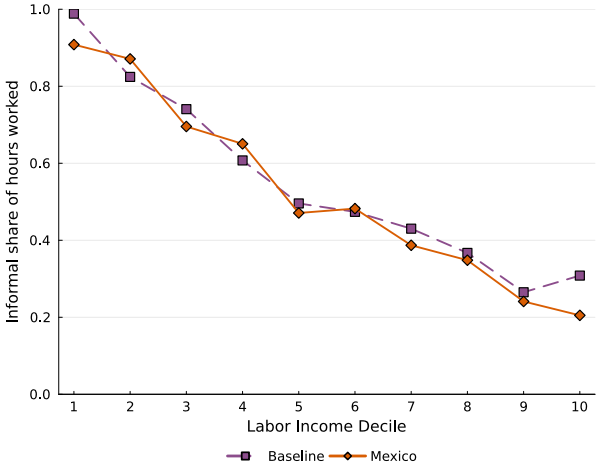
Transfers provide income to low-ability households and insurance against formal-job loss. Rais-



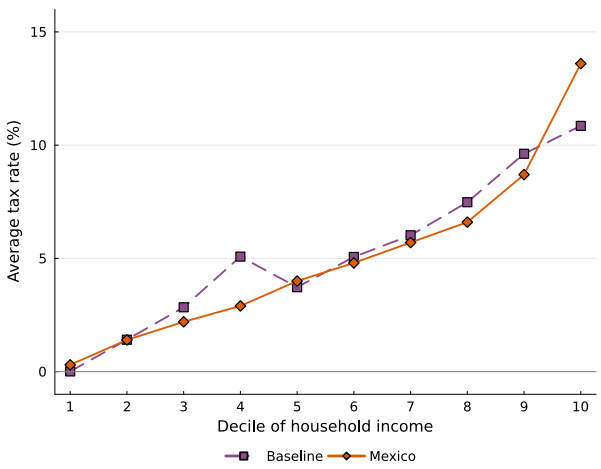
Notes: In the homogeneous economy every formal job separates at rate 0.169, the baseline average, and every household without a formal job finds one at rate 0.191, the equilibrium job-finding rate of this economy; no parameter is recalibrated. Panel (a) reports the informal share of hours worked, and panels (c) and (d) the transition rates, by labor-income decile. Panel (b) reports cumulative distributions of productivity, h^s for households without a formal job and h for households with one, each scaled to its group's population share.

Figure D.3: Informality rate, productivity distribution, and formal-sector transition rates

(a) Informality, share of hours worked



(b) Average tax rates, including the profit tax



Notes: Panel (a) reports the informal share of hours worked, by decile of the worker's own labor income, in the baseline economy and in Mexico's National Survey of Occupation and Employment (ENOE, 2024). Panel (b) reports average tax rates, in percent, by decile of household income. Baseline ($\tau^P = 0.015$): labor tax plus the tax on business profits, $\tau^{\text{profit}}\,\Pi(h)$, divided by total gross household income (labor, capital, and profit income); the tax on capital income, $\tau^k ra$, is not included. Mexico: personal income tax paid by individuals (*ISR personas físicas*) divided by total gross household income, by decile of households, as estimated by the SHCP from ENIGH 2024 in the scenario that assigns the tax only to earners affiliated with social security (same report as Panel 6d).

Figure D.4: Informal share of hours and average tax rates including the profit tax, by decile

Table D.3: Effects of progressive fiscal policy: comparative statics

	Optimal policy		Welfare	Ex-post Gini	Informality		
	τ^P	τ^L			Average	Bottom 25%	Top 25%
Baseline	0.24	0.834	-0.839	0.468	0.542	0.837	0.259
A: Government expenditure to GDP ratio, G/Y (Baseline 0.15)							
0.10	0.28	0.893	-0.782	0.431	0.494	0.746	0.269
0.20	0.20	0.762	-0.899	0.501	0.606	0.920	0.259
B: Unemployment risk, q (Baseline 0.06)							
0.01	0.31	0.879	-0.701	0.396	0.400	0.705	0.111
0.10	0.21	0.803	-0.914	0.504	0.625	0.898	0.344
C: Lump-sum transfers as a fraction of GDP, τ^T/Y (Baseline 0.0)							
0.01	0.23	0.821	-0.830	0.467	0.550	0.857	0.250
0.02	0.22	0.808	-0.823	0.463	0.560	0.876	0.242

Notes: Each row reports outcomes at that economy's welfare-maximizing progressivity τ^P. The tax scale τ^L balances the budget, financing the specified government consumption and transfers. Welfare is population-average expected utility, including labor disutility, in levels. The ex-post Gini measures labor-earnings inequality. Gini coefficients and informality rates are fractions. Other structural parameters remain at the static-model baseline.

ing transfers to 1% or 2% of GDP increases welfare, lowers optimal progressivity slightly, and makes earnings more equal.

E Details and Additional Results of the Static Model

E.1 Equilibrium Conditions

The government levies taxes on formal earnings to finance spending G and lump-sum transfers τ^T. Its balanced budget is

$$G + \tau^T = (1-q)\int_0^\infty \left(w^f hn\varpi - \tau^L (w^f hn\varpi)^{1-\tau^P}\right) f_h(h)\, dh, \tag{E.1}$$

where f_h is the density of ability. In the baseline experiments, $\tau^T = 0$ and τ^L adjusts to keep $G/Y = 0.15$.

Definition 1 *An equilibrium consists of allocations* $\{n, \varpi, c^f, c^s, L^s, L^f, Y\}$ *and prices* $\{w^f, w^s\}$ *such that:*

- *Households choose* n *and* ϖ *to maximize utility given wages, the tax schedule* (τ^L, τ^P)*, transfers* τ^T*, and unemployment risk* q*.*
- *Firms maximize profits given wages.*
- *The government keeps a balanced budget.*
- *Markets clear:*

$$Y = (1-q)\int_0^\infty c^f f_h(h)\, dh + q\int_0^\infty c^s f_h(h)\, dh + G, \tag{E.2}$$

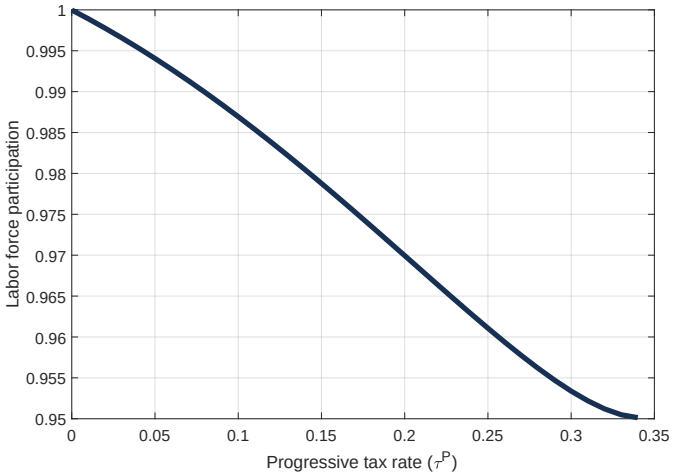
$$L^f = (1-q)\int_0^\infty hn\varpi f_h(h)\, dh, \tag{E.3}$$

$$L^s = \int_0^\infty \Xi h^\xi n(1-\varpi) f_h(h)\, dh. \tag{E.4}$$

The net marginal return to reallocating labor toward formality, conditional on finding a job, is

$$\tilde{w} = (1-\tau^P)\tau^L (w^f hn)^{1-\tau^P} \varpi^{-\tau^P} - w^s \Xi h^\xi n. \tag{E.5}$$

At an interior allocation, equation (7) equates $(1-q)\tilde{w}/c^f$ to the expected marginal utility loss $qw^s \Xi h^\xi n/c^s$ when no formal job is found.



Notes: Static-model equilibria as tax progressivity varies. Lump-sum transfers are zero, and the tax scale adjusts to finance government spending of 15% of output. Other structural parameters are held fixed.

Figure E.1: Progressivity and total labor supply

E.2 Additional Fiscal-Policy Results

Figure E.1 complements the welfare and informality profiles in Figure 5. Total labor supply responds relatively little to progressivity: increasing τ^P from zero to 0.35 reduces labor supply by about 5%. Most adjustment therefore occurs through the allocation of work across sectors.

F Additional Empirical Evidence

F.1 Survey Coverage and Earnings Inequality

The four surveys are carried out continuously over the year. The number of households interviewed was 708,340 in Brazil, 277,158 in Colombia, 496,153 in Mexico, and 33,886 in Peru. Each survey is representative at the national, urban, and rural levels and is an official source of labor-market statistics. For Mexico, the official source of income inequality statistics is the National Household Income and Expenditure Survey (ENIGH), which provides more detailed information on income sources but less information on labor-market outcomes than the ENOE.

Brazil and Mexico interview households for five consecutive quarters. Peru follows a subsample annually for up to five years, while Colombia includes retrospective labor-market questions. Appendix A describes the surveys and the definitions used in the cross-country comparisons.

Table F.1 reports inequality in household per-capita labor earnings. Inequality is substantial in all four countries, and the percentile ratios show that the bottom of the distribution lies particularly far below the median. The Gini ranges from 0.424 in Mexico to 0.580 in Colombia.

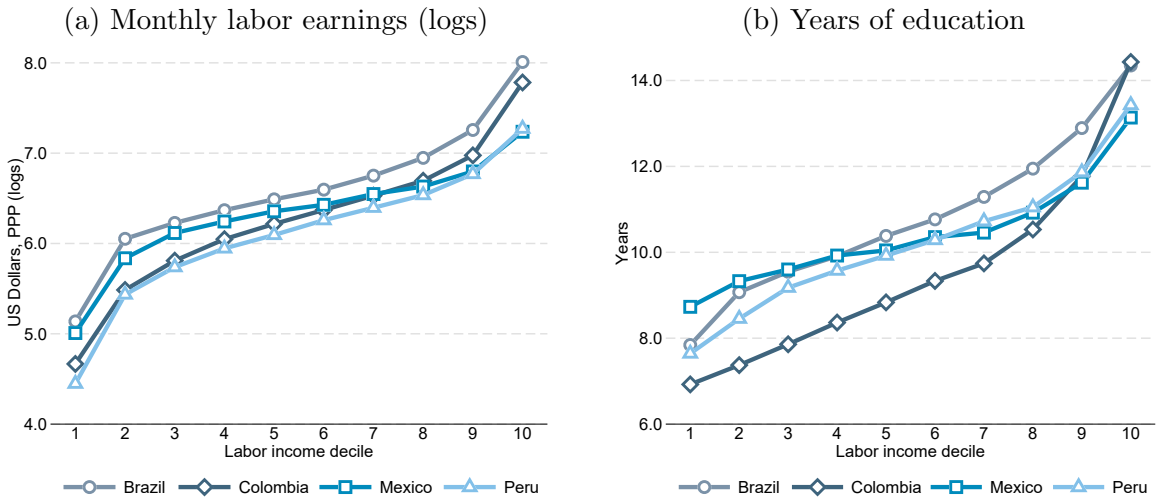
	Brazil	Colombia	Mexico	Peru
Gini	0.543	0.580	0.424	0.498
Variance of logs	1.145	1.303	0.815	1.164
Percentile ratio 90/10	13.200	16.833	7.998	13.339
Percentile ratio 75/25	3.500	4.127	2.907	3.463
Percentile ratio 90/50	3.451	4.040	2.544	3.159
Percentile ratio 10/50	0.261	0.240	0.318	0.237
Survey year	2024	2023	2024	2023

Notes: Household per-capita labor earnings equal total household labor earnings divided by household size. The sample includes households with positive labor earnings. Earnings cover the main job in Colombia and Mexico and all jobs in Brazil and Peru; Peru's annual earnings are divided by 12. Estimates use survey weights. Appendix A describes the surveys.

Table F.1: Inequality measures, households' per capita labor earnings

F.2 Earnings and Education Across the Income Distribution

Figure F.1 shows monthly labor earnings in logs and average years of education by labor-income decile. Earnings are measured in PPP-adjusted US dollars. Earnings differences are largest at the



Notes: The sample covers individuals aged 15 and older in households with positive labor earnings, grouped into country-specific deciles of household per-capita labor earnings. Panel (a) shows employed individuals' monthly earnings in logs, measured in PPP-adjusted US dollars for the survey year; panel (b) shows average completed years of schooling. Earnings cover the main job in Colombia and Mexico and all jobs in Brazil and Peru. Survey years are 2024 for Brazil and Mexico and 2023 for Colombia and Peru. Estimates use survey weights; Appendix A describes the surveys.

Figure F.1: Monthly earnings (logs) and average years of education by labor income decile, 2023/24

ends of the distribution. Consistent with its lower earnings inequality in Table F.1, Mexico displays a flatter gradient and lower earnings in the top decile than the other countries.

Schooling rises with income in all four countries. Colombia has lower average schooling (9.1 years), but schooling in its top decile is comparable with Brazil and Peru. As with earnings, the schooling gradient in Mexico is flatter, including at the top.

F.3 Additional Labor-Market Comparisons

The unemployment gradients in Figure 1 differ across countries, with larger differences between the bottom and top in Brazil and Colombia. These comparisons use individuals aged 15 and older in households with positive labor earnings, ranked by household per-capita labor earnings. The employment rate is the employed-to-population ratio, the unemployment rate is the unemployed-to-labor-force ratio, and the informality rate is the informal-to-employed ratio.

G Details of the Quantitative Model

This appendix supplies the factor-price equations, equilibrium definition, and income-process annualization for Section 4.

G.1 Factor Prices and Aggregation

Under perfect competition, net of depreciation δ, rental rate of capital r and the price of intermediate inputs p^H are given by:

$$r + \delta = \alpha\nu\frac{Y}{K}, \quad p^H = (1-\alpha)\nu\frac{Y}{H}. \tag{G.1}$$

From (10), the relative price of formal intermediate inputs p^f and the return per efficiency unit of informal labor w^s satisfy:

$$p^f = (1-\alpha)\nu\frac{Y}{H}\left(\frac{H}{\Omega L^f}\right)^{1/\varepsilon}, \tag{G.2}$$

$$w^s = \varkappa p^f\left(\frac{\Omega}{\varkappa}\cdot\frac{L^f}{L^s}\right)^{1/\varepsilon}. \tag{G.3}$$

Aggregation. Combining (9) and (10), we can express the aggregate production function as:

$$Y = \left\{A\left[(\Omega(1-\lambda^s))^{\frac{\varepsilon-1}{\varepsilon}} + (\varkappa\lambda^s)^{\frac{\varepsilon-1}{\varepsilon}}\right]^{\frac{\varepsilon\nu(1-\alpha)}{\varepsilon-1}}\right\}\left(K^\alpha L^{1-\alpha}\right)^\nu, \tag{G.4}$$

where $L = \left(L^f + L^s\right)$ and measured TFP depends on the informal share in efficiency units $\lambda^s \equiv \frac{L^s}{L}$.

Equation (G.4) makes the model's TFP–informality link explicit: measured TFP is

$$\mathrm{TFP}(\lambda^s) \;=\; A\left[\left(\Omega(1-\lambda^s)\right)^{\frac{\varepsilon-1}{\varepsilon}} + \left(\varkappa\lambda^s\right)^{\frac{\varepsilon-1}{\varepsilon}}\right]^{\frac{\varepsilon\nu(1-\alpha)}{\varepsilon-1}}. \tag{G.5}$$

For $\varepsilon > 1$, this expression decreases with λ^s when $\lambda^s > \varkappa^{\varepsilon-1}/(\Omega^{\varepsilon-1} + \varkappa^{\varepsilon-1})$. Reallocation changes measured TFP through the composition of production; Section 5 reports its magnitude across the policy experiments.

G.2 Stationary Equilibrium

A stationary equilibrium consists of household value and policy functions $\{V, g^a, g^\ell, g^\kappa, g^c\}$ over states (a, h, F), aggregate inputs $\{K, L^s, L^f\}$, and prices $\{r, p^H, p^f, w^s, w^f\}$. It also includes the

labor agency's value $J(L^f)$ and vacancies V, the job-finding rate f, labor-market tightness θ, fiscal instruments $\{\tau^c, \tau^k, \tau^{\text{profit}}, \tau^L, \tau^P, \tau^T\}$, and an invariant household distribution $\mu(a, h, F)$, such that:

1. households solve their optimization problem given prices, labor market transition rates, and fiscal tools,
2. factor prices satisfy (G.1), (G.3), (G.2), and (20),
3. the zero-profit condition for the labor agency (19) holds,
4. the government budget satisfies (21),
5. the distribution $\mu(a, h, F)$ is stationary and satisfies all agents' policy functions and aggregate consistency conditions,
6. all markets clear.

Market clearing. The formal-hours share is $g^\kappa(a, h, 0) = 0$ for households without formal job access. Market clearing requires:

$$\begin{aligned} K &= \int g^a(a, h, F)\, d\mu(a, h, F), \\ L^f &= \int g^\kappa(a, h, F) g^\ell(a, h, F) \cdot h^f \, d\mu(a, h, F), \\ L^s &= \int \left[1 - g^\kappa(a, h, F)\right] g^\ell(a, h, F) \cdot h^s \, d\mu(a, h, F), \\ Y &= \int \frac{\Pi(h)}{1 - \nu} \, d\mu(a, h, F) \end{aligned} \tag{G.6}$$

G.3 Annualization of the Income Process

Income-process parameters are annualized from quarterly values $\rho_q = 0.995$ and $\sigma_q^2 = 0.01$. The corresponding annual persistence is $\rho = \rho_q^4 = 0.980$, and the annual innovation standard deviation is $\sigma_h = \sigma_q \sqrt{1 + \rho_q^2 + \rho_q^4 + \rho_q^6} = 0.199$. These values imply long-run dispersion $\sigma_{LR} = \sigma_h / \sqrt{1 - \rho^2} = 1.00$.